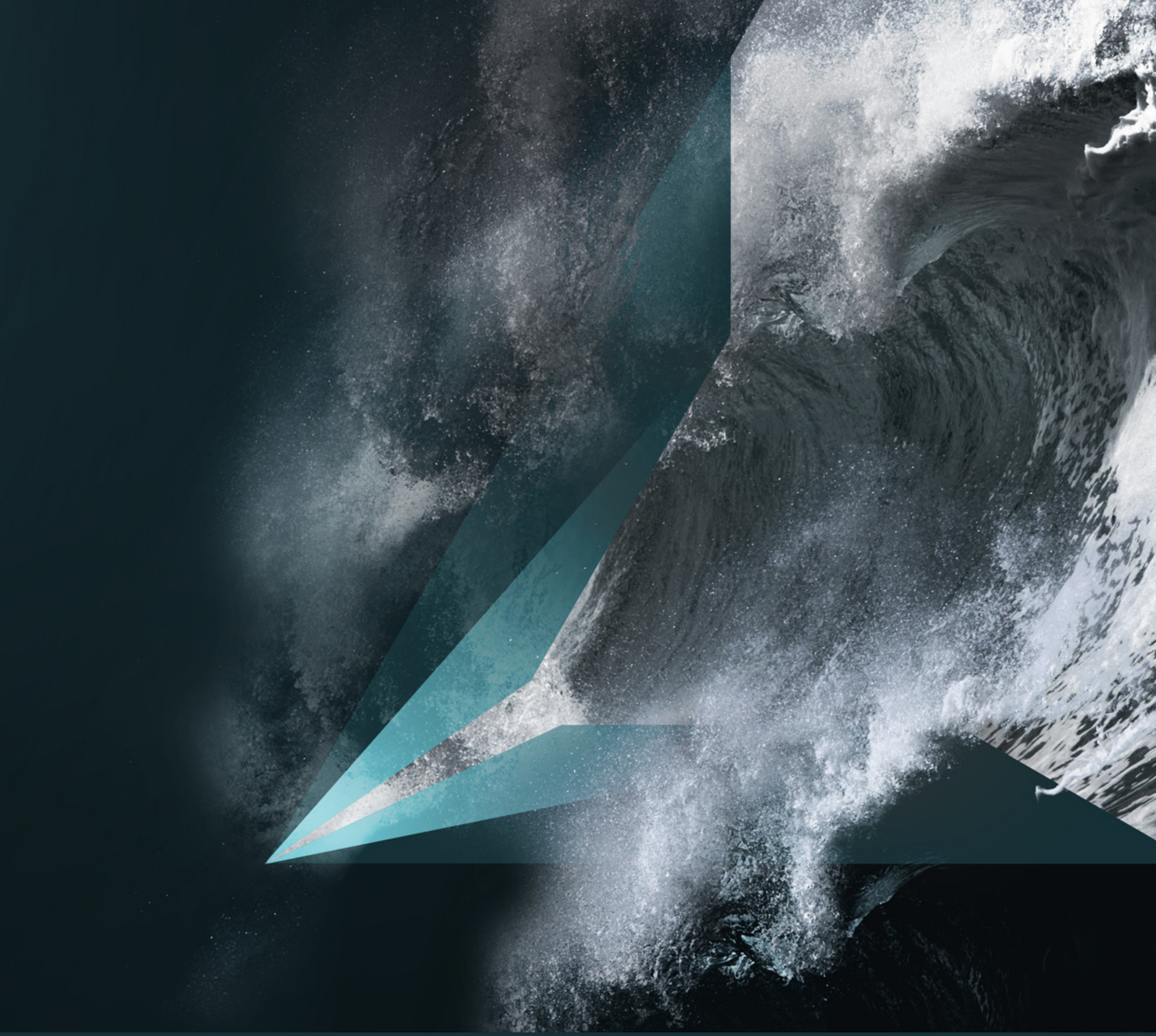

SWIFT SURVIVOR RECOVERY SYSTEM

Maintenance Manual

Document number PN/SWI/011-00

Version 2.0

ZELIM

MANAGEMENT CONTROL

DATE	DOCUMENT REFERENCE	VER NO.
08/08/24	SWIFT Maintenance Manual PN/SWI/011-00	1.0
07/01/25	SWIFT Maintenance Manual PN/SWI/011-00	1.1
11/03/26	SWIFT Maintenance Manual PN/SWI/011-00	2.0

	NAME	POSITION	COMPANY	DATE	SIGNED
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APPROVER	S. Gregory	COO	Zelim	11/03/26	

PERMANENT REVISION LIST

Revisions or amendments embodied in this manual, that are certified by an approved organisation, other than that applicable to this initial certification must be recorded on separate record sheets.

DATE	COMMENT	BY	VER NO.
07/01/25	Minor content updates and formatting improvements.	P. Lloyd	1.1
11/03/26	Full document rewrite. Updated structure, maintenance procedures, technical specifications and formatting.	P. O'Mara	2.0

LIST OF SERVICE BULLETINS

On receipt of a Service Bulletin (SB) that is associated with this manual, maintenance personnel may hand amend this list to record the receipt of the SB. Once the contents of the SB are incorporated into the manual it will be permanently added to this list.

SB NO.	DATE	COMMENT	INCORPORATED	
			ISSUE	DATE
001	19/08/25	Modification of the SWIFT control panel to include a Type B three-pole RCD in-line with the motor circuit.		

LIST OF SERVICE NEWSLETTERS

On receipt of a Service Newsletter (SN) that is associated with this manual, maintenance personnel may hand amend this list to record the receipt of the SN. Once the contents of the SN are incorporated into the manual it will be permanently added to this list.

SN NO.	DATE	COMMENT	INCORPORATED	
			ISSUE	DATE

CONTENTS

1 INTRODUCTION

This chapter defines the purpose and structure of the SWIFT Maintenance Manual.

1.1 General

A This manual contains instructions for the SWIFT product and shall be used by individuals responsible for performing maintenance on this equipment.

B Details of disassembly, cleaning, testing, repair, assembly, storage, disposal, and spare parts are provided in this manual.

C Maintenance procedures shall only be carried out by Zelim and/or an endorsed maintenance organisation.

D A record of all maintenance activities shall be maintained by maintenance personnel and retained for the specified period by Zelim and/or the endorsed maintenance organisation.

E Zelim is the Design Authority for the SWIFT. The Company has developed this equipment and its associated regulatory procedures to ensure reliable operation under all climatic, weather, and operational conditions. While comments intended to improve the efficiency of maintenance procedures are always welcome, no changes to these procedures may be made without Zelim’s prior approval. Unauthorised modifications may cause the equipment to malfunction and may also void the equipment’s approval.

F This manual is the copyright of Zelim Limited. It is supplied in confidence and shall not be used for any purpose other than that for which it is provided. This manual shall not be reproduced, in whole or in part, without the prior permission of Zelim Limited.

1.2 Structure of the manual

A Chapters and page numbers

1 Chapters are assigned to each main topic within this manual.

2 Auxiliary sections, such as appendices, are also included.

B Record of revisions

1 All issues of this manual are recorded in the Permanent Revision List.

2 Any changes from the previous issue shall be shown in **red text**. The start and end of a change will be marked by red pointers > <. Text deleted from the previous issue will also be indicated by start and end red pointers.

3 All amendments shall be summarised in Section 1.6 – List of Changes.

1.3 Warnings, cautions and notes

WARNING A warning calls attention to a procedure which if incorrectly performed is liable to cause injury or death to personnel.

CAUTION A caution calls attention to a procedure which if incorrectly performed is liable to cause damage to the equipment or its components.

NOTE A note calls attention to methods.

1.4 Health and safety

A Personnel responsible for the maintenance of the SWIFT shall ensure that adequate safety precautions are observed during all maintenance and related activities.

B Zelim shall not, by virtue of these instructions, be deemed to have assumed any responsibilities of the Service Agent or Operator under the Health and Safety at Work etc. Act 1974 or any related legislation.

C Materials used during maintenance of this product may be hazardous. Food, beverages, cups, and other eating or drinking utensils must not be stored in areas where such materials are handled or stored. Hands must be washed thoroughly with soap and water immediately after maintenance work and before eating or drinking.

1.5 Acronyms and abbreviations

AC. / ABRV.	MEANING
A/R	As required
Approx	Approximately
cm	Centimetre
EP	Extreme pressure
Fig	Figure
IAW	In accordance with
In situ	In situation
Intro	Introduction
IP	Ingress Protection rating
kg	Kilogram
LM	Locally manufactured

AC. / ABRV.	MEANING
LP	Locally purchased
mm	Millimetre
MOB	Man overboard
m/s	Metres per second
No	Number
Qty	Quantity
Rev	Revision
SB	Service bulletin
SN	Service newsletter
TBA	To be agreed

1.6 List of changes

CHAPTER	PAGE	DESCRIPTION	REASON

2 LEADING PARTICULARS

This chapter provides the principal specifications and key identifying information for the SWIFT. It details the main dimensions, weights, performance characteristics, and other data required for reference during maintenance and operation.

2.1 SWIFT reference views

A Typical views of the SWIFT assembly and major components are illustrated in Figures 1 to 4. These figures show representative orientations of the unit for identification and reference purposes only.

Fig 1 SWIFT front view

Fig 2 SWIFT side view (motor side)

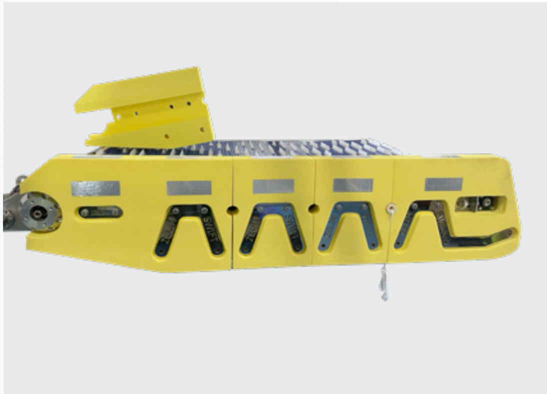

Fig 3 SWIFT side view (non-motor side)

Fig 4 SWIFT rear view

2.2 SWIFT technical particulars

Dimensions

Length	**1986 mm**
Width	**1739 mm**
Height	**420 mm**

Weight

SWIFT	**≈ 175 kg**

Inputs

Power supply	**230 V AC, 50 Hz, 0.75 kW**

Other

Safe working load	**137 kg**
Recovery angle	**≤ 40°**

2.3 Shipping crate particulars

Dimensions

Length	**2060 mm**
Width	**1860 mm**
Height	**760 mm**

Weight

Empty shipping crate	**≈ 160 kg**
SWIFT assembly packed in shipping crate	**≈ 335 kg**

2.4 Manual winch particulars

A SWK-LB-250-4-20M (obsolete)

Hand-operated stainless steel winch with automatic ratchet stop in both directions.

Dimensions

Length	**232 mm**
Width	**155 mm**
Height	**175 mm**

Weight	**≈5.6 kg**

Other

Rope supplied	**8 mm Dyneema®** [1]

B SWK-LB-650-6-20M

Hand operated stainless steel winch with automatic ratchet stop in both directions.

Dimensions

Length	**273 mm**
Width	**192 mm**
Height	**210 mm**

Weight	**≈9 kg**

Other

Rope supplied	**8 mm Dyneema®** [1]

1 Rope length is determined during commissioning to suit the vessel specification.

3 MAINTENANCE SCHEDULE

This chapter defines the maintenance requirements for the SWIFT, including planning responsibilities for scheduled maintenance, maintenance intervals, and certification following maintenance. All maintenance shall be performed in accordance with the instructions contained in this manual.

A Scheduled maintenance planning (annual and 30-month) shall be the responsibility of the operator. The operator shall pre-book maintenance directly with Zelim and/or an endorsed maintenance organisation. Any operation of the SWIFT with overdue scheduled maintenance shall be at the client's risk and may invalidate product warranty and certification.

B Maintenance shall be carried out by Zelim and/or an endorsed maintenance organisation and may require the SWIFT to be removed from the vessel. Unscheduled maintenance may be required following a fault, damage, abnormal operation, or as directed by Zelim.

C Once satisfied that the SWIFT is safe and operational following annual or 30-month depth maintenance, Zelim shall issue a SWIFT Recovery System Maintenance Certificate. Following unscheduled maintenance, the Unscheduled Maintenance Record (Appendix C) shall be completed.

3.1 Maintenance matrix

The SWIFT shall be maintained in accordance with the outlined maintenance matrix (Table 1).

Table 1 Maintenance matrix

MAINTENANCE ACTIVITY	CONDUCTED BY	IN SITU OR REMOVED	PAPERWORK
Annual maintenance	Zelim/ endorsed maintenance organisation	In situ or removed	Annual Maintenance Record (Appendix A) completed and certificate issued by Zelim
30-month depth maintenance	Zelim/ endorsed maintenance organisation	Removed	30-month Depth Maintenance Record (Appendix B) completed and certificate issued by Zelim
Unscheduled maintenance	Operator/ Zelim/ endorsed maintenance organisation	In situ or removed	Unscheduled Maintenance Record (Appendix C) completed

4 DISASSEMBLY

This chapter provides the procedures for disassembling the SWIFT and its components for inspection, maintenance, or repair. All disassembly operations shall be carried out in accordance with these instructions to prevent damage and ensure safe reassembly.

4.1 Equipment

Disassembly of the SWIFT shall be carried out using the equipment specified in Table 2.

Table 2 Disassembly equipment

NOMENCLATURE	PART NUMBER
Mechanical lifting winch	Locally manufactured / locally purchased
Flat work bench	Locally manufactured / locally purchased

4.2 Removal of SWIFT from vessel

1. Remove the SWIFT from the vessel in accordance with the SWIFT Survivor Recovery System Installation Guide (PN/SWI/010-00).

4.3 Removal of SWIFT from shipping crate

1. Attach mechanical lifting equipment to the two lifting eyes.
2. Lift the SWIFT out of the shipping crate.
3. Place the SWIFT on a stable working platform.

4.4 Pre-disassembly

1. Complete the pre-disassembly inspection in the 30-Month Depth Maintenance Record (Appendix B).

CAUTION Remove the SWIFT from the shipping crate (if applicable) using suitable lifting equipment and place on a stable working platform.

2 Photograph and record the condition of the SWIFT (including a clear photo of the data plate showing serial number) and any returned components prior to disassembly.

3 Conduct a functional test of the SWIFT to confirm its condition, in accordance with Chapter 7.1 – Functional testing.

4 Clean the SWIFT thoroughly prior to maintenance, in accordance with Chapter 5 – Cleaning.

4.5 Disassembly

4.5.1 Hinge mounting bracket

A Hinge bracket removal

1 If the hinge mounting bracket (Fig 5) is attached to the SWIFT, remove it by loosening the four M16 stainless steel bolts (Fig 6) connecting the hinge bearings (Fig 7) between the brackets and stub axles on the chassis.

Fig 5 Hinge mounting bracket

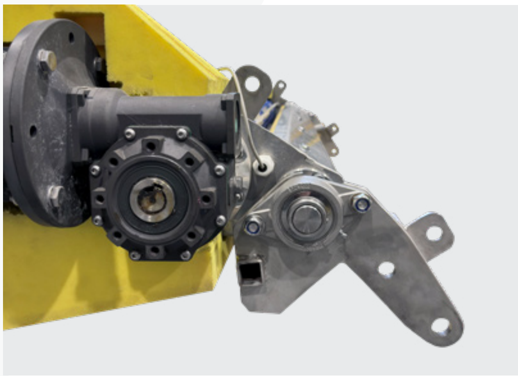

Fig 6 Hinge mounting bracket M16 stainless steel bolts

Fig 7 Hinge bearing

4.5.2 Floatation foam moulds

A Floatation foam moulds removal

1 Using mechanical lifting equipment, attach suitable slings to the detachable lifting eyes.

NOTE To prevent damage to the foam moulds from chains or slings, use sacrificial pieces of foam.

2 Raise the SWIFT and rest it on the motor and gearbox end (Fig 8).

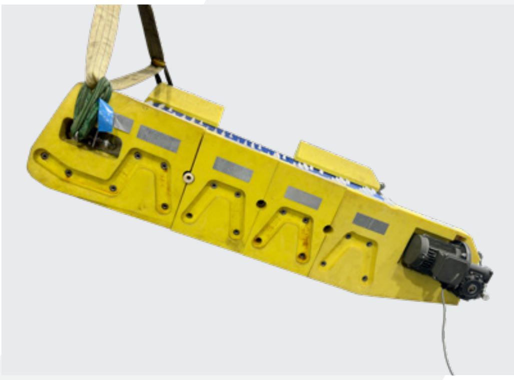

Fig 8 SWIFT raised, resting on motor and gearbox end

3 Remove bolts, washers and clamp plates from both sides (Fig 9).

Fig 9 Clamp plates

4 Remove the 2 × foam moulds No. 1 from each side of the chassis (Fig 10).

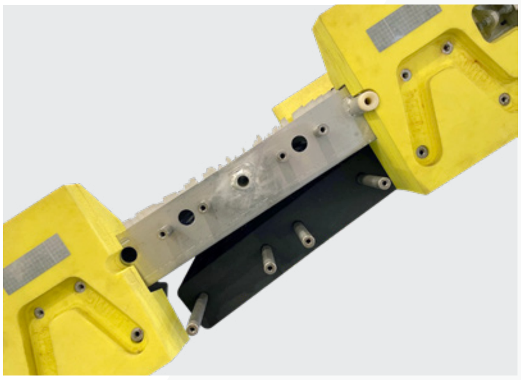

Fig 10 Foam moulds No. 1 removed

5 Remove 4 × tie bars.

6 Lower the SWIFT onto a flat surface and remove the chains/slings.

7 Remove foam mould No. 4 or No. 5 from the end of the chassis (Fig 11).

Fig 11 Foam mould No. 4 at the end of the chassis

8 Remove the 2 × tie bars and 12 × black and yellow foam moulds (8) (Fig 12).

Fig 12 Black and yellow foam moulds

9 Remove the other foam mould 4 or 5 and 2 tie bars.

10 Attach chains/slings to the two lifting eyes on the motor end and raise the SWIFT using mechanical lifting equipment.

11 Remove foam mould (3).

12 Carefully remove foam mould (2) from around the drive motor and gearbox and 2 × tie bars.

13 The tie bars and 12 × pieces of undercarriage black foam mould (6) will detach and remain on the flat surface.

14 Separate the tie bars from the undercarriage black foam moulds.

15 Lower the chassis onto the flat surface.

4.5.3 Conveyor belt

A Conveyor belt removal

1 Loosen the 4 × galvanised M12 half nuts from both sides of the chassis (Figs 13 and 14).

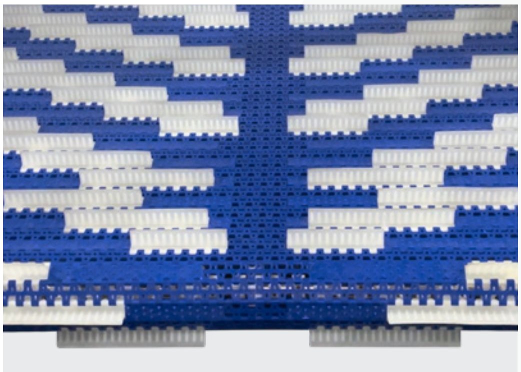

Fig 13 Conveyor belt

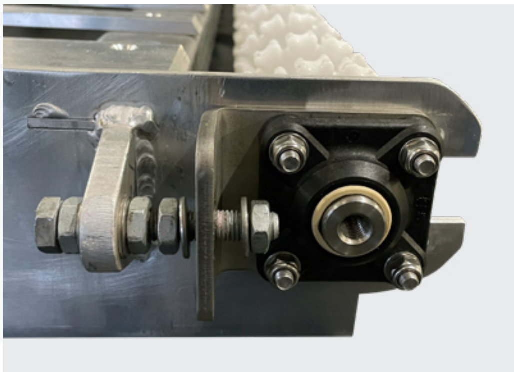

Fig 14 Galvanised half nut

2 Loosen the 4 × M8 Nyloc nuts securing the tension shaft bearing on both sides of the chassis (Fig 15).

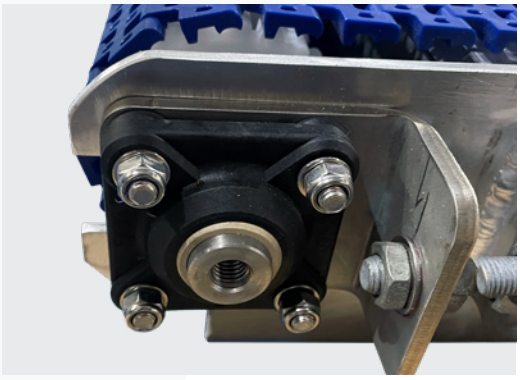

Fig 15 Nyloc nuts

3 To split the conveyor belt, lift the edge and identify a single white 5 mm diameter plastic rod connecting the modular system. Remove the white plastic rod by gripping with long nose pliers, (Fig 16) and pull out as per (Fig 17).

NOTE To ease removal, pull the plastic rod from a position as close to the motor as practicable.

Fig 16 Identify single white plastic rod

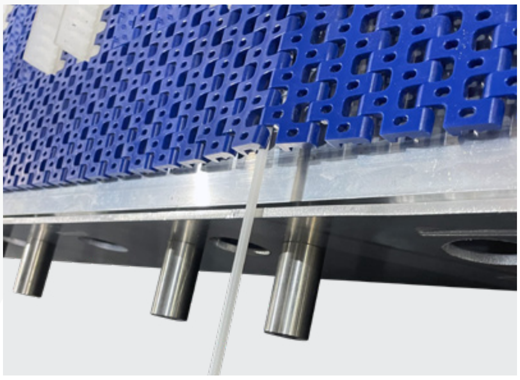

Fig 17 White plastic rod removal

4 Remove the conveyor belt using a ratchet spanner with a 24 mm socket on the drive nut at the end opposite the motor (Fig 18) and manually turning anticlockwise. Roll up the conveyor as you go along until it becomes free.

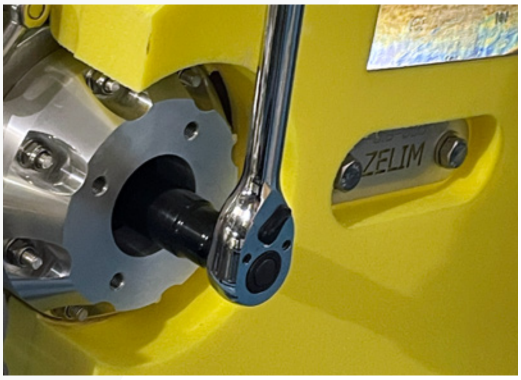

Fig 18 Ratchet spanner with 24 mm socket

4.5.4 Detachable lifting eyes

A Detachable lifting eyes removal

1 To remove the two detachable lifting eyes (Fig 19) from the chassis, unscrew them counter-clockwise.

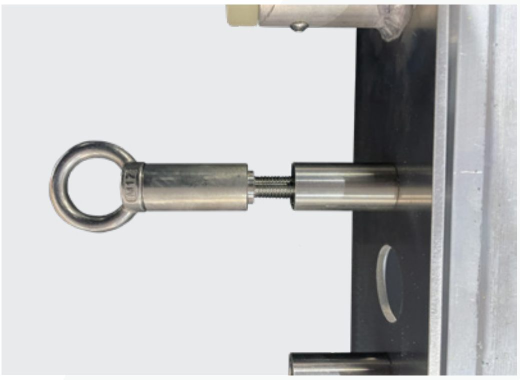

Fig 19 Detachable lifting eyes

5 CLEANING

This chapter details the procedures and equipment required to clean the SWIFT and its components. All cleaning shall be performed using the specified materials and methods to prevent damage and ensure continued operational reliability.

5.1 Equipment

The SWIFT and its components shall be cleaned using the equipment specified in Table 3.

Table 3 Cleaning equipment

NOMENCLATURE	PART NUMBER
Mild detergent suitable for marine environment	Locally purchased
Lint free cloth	Locally purchased
Sponge	Locally purchased
Soft brush	Locally purchased

5.2 Maintenance and storage

The following cleaning process shall be used to clean the SWIFT and its components when removed from a vessel for maintenance or storage.

1. Rinse the SWIFT with clean water.
2. Using a mixture of clean water and mild detergent, brush down the entire assembly to remove any debris and contamination.
3. Rinse with clean water and allow the SWIFT to dry naturally.

CAUTION Pressure washers must not be used when cleaning the SWIFT. Damage may be caused to the foam parts, motor and the electrical system.

CAUTION Bleach must not be used when cleaning SWIFT. Damage may be caused to the foam parts and the conveyor belt.

5.3 Annual and 30-month depth maintenance cleaning

The following cleaning process shall be used to clean the SWIFT and its components during annual and 30-month depth maintenance.

5.3.1 Hinge mounting bracket

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.2 Floatation foam moulds

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.3 Conveyor belt

1. Hand wash using a sponge / soft brush and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.4 Tension shaft assembly

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.5 Motor and gearbox

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.6 Drive shaft assembly

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.7 Bearing mounting brackets and drive shaft bearings

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.8 Stub pivot axles

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.9 Tape switch

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.10 Chassis

1. Hand wash using a sponge / soft brush and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.11 Nylon fairlead

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.12 Winch (if returned)

1. Hand wash using a sponge and water mixed with a mild detergent.
2. Rinse with clean water and allow the component to dry naturally.

5.3.13 Electrical box, 3-push-button control box and emergency stop button

1. Clean the outer casing and electrical cables using a damp, lint-free cloth.
2. Allow the components to dry naturally.

6 INSPECTION AND MAINTENANCE PROCEDURES

This chapter describes the inspection and maintenance requirements for the SWIFT, including annual and 30-month depth maintenance intervals. All work shall be carried out by Zelim or an endorsed maintenance organisation in accordance with this manual.

6.1 Annual maintenance (12 months)

6.1.1 Overview

A Annual maintenance may be conducted in situ. The SWIFT may be removed from the vessel if required.

NOTE If removal from the vessel is required, refer to Section 6.2.2 – SWIFT removal and transportation.

B Any components found to be unserviceable shall be repaired or replaced in accordance with Chapter 8 – Repair and Replacement, or cleaned in accordance with Chapter 5 – Cleaning.

NOTE All electrical maintenance, repair, or component replacement shall be undertaken by a suitably qualified marine engineer.

C When conducting annual maintenance, record all checks completed and note any observations in Appendix A – Annual Maintenance Record.

D As part of annual maintenance, Zelim and/or an endorsed maintenance organisation shall review the customer's monthly maintenance records.

E Any discrepancies between the records and the condition of the product that may affect the product warranty shall be reported to Zelim immediately.

F The maintenance organisation shall ensure that all manuals and relevant documentation are maintained at the current revision level.

G Once satisfied that the SWIFT is safe and operational, Zelim and/or the endorsed maintenance organisation shall issue a SWIFT Recovery System Annual Maintenance Certificate, valid for 12 months.

6.1.2 Annual pre-maintenance checks

1. The maintenance organisation is to review all customer maintenance paperwork for its history and any existing faults on the SWIFT since the last maintenance.
2. Any immediate concerns should be raised with Zelim as required and/or if the product warranty is in question.

3. Prior to maintenance, take clear photographs of the SWIFT to document its condition, including any signs of wear, corrosion, damage, or marine growth. Include a clear photo of the SWIFT data plate showing the serial number.

   NOTE The data plate is located at the end of the chassis, adjacent to the motor.

6.1.3 Annual maintenance process

1. Take clear photos of the SWIFT in situ and the surrounding installation area, and store them locally as part of the maintenance records.
2. Confirm all monthly inspection records are present, complete, and signed.
3. Conduct a visual inspection of the SWIFT and accessories, noting any corrosion build-up, debris accumulation, or encrustation by marine growth.
4. Remove, inspect, and lubricate mounting feet hinge pins (if fitted) with lithium EP2 grease.
5. Using a grease gun with lithium EP2 grease, lubricate the mounting bracket bearings via the grease nipples.
6. Check the security of all accessible structural and mechanical fasteners. Re-torque as required in accordance with the specified values in Table 4 – Bolt torques and spanner sizes (Section 8.3 – Bolt and nut torque requirements).
7. Check that the sprockets on the drive and tension shafts are correctly aligned with the conveyor belt.
8. Visually verify conveyor belt tracking and tension while running. Ensure the belt runs centrally on the sprockets without deflection, flutter, or vibration. Adjust belt tension only if visible slackness or misalignment is observed.
9. With the conveyor belt running, confirm smooth operation and ensure there are no squeaks, rubbing, or signs of heat build-up at the shaft bearings.
10. Conduct an operational speed check on the conveyor in both forward and reverse directions by marking a point on the belt and recording the time for one complete rotation to confirm consistent drive speed. Repeat three times each direction and record the average. This check confirms drive consistency and identifies early signs of motor degradation, gearbox wear, or increased system friction.
11. IAW Chapter 7.1, complete a functional test.
12. Inspect the motor and gearbox for lubrication leaks or signs of oil contamination.
13. Deploy the SWIFT and observe for smooth, controlled operation. Observe winch behaviour during deployment.
14. With the SWIFT supported by buoyancy, fully unspool the winch rope. Inspect the rope along its full length and confirm correct routing through the nylon fairleads and SWIFT chassis.

15 Check the rope is securely retained at the winch rope clamp and at the vessel strong point. Ensure the rope is retained by both clamping screws and torque the M8×12 rope clamp screws to 16 N·m (dry, no Loctite).

16 Inspect the rope for fraying, furring, thinning, glassing (heat damage), dirt, or oil contamination.

17 Check for zinc plating loss on the winch drum side cheeks where the rope makes contact.

18 Inspect the nylon fairleads for signs of wear where the rope passes through the SWIFT chassis.

19 Ensure the winch is adequately lubricated. Apply lithium EP2 grease to the handle pivot, drum bearings, and any exposed gear or pawl interfaces as required.

CAUTION Avoid contaminating the winch drum or rope with grease.

20 Operate the winch to pay out and recover the rope, confirming smooth engagement, no slipping, and positive ratchet/brake holding in both directions.

21 Isolate power and inspect the control box for water ingress, corrosion, or loose connections.

22 Inspect cables and terminations, then confirm correct operation after restoring power.

23 Conduct a trial recovery of an adult MOB manikin, recording current draw under load IAW Chapter 7.1.4 (Current draw testing). Log all readings.

WARNING Task to be carried out by a competent person trained in the safe use of a clamp meter and the SWIFT recovery system. No covers are to be removed and no electrical connections disturbed. If access requires opening the control box or exposed live parts are present, an electrically competent technician/electrician shall perform the test.

24 Record any additional observations, defects, or actions.

6.2 30-month depth maintenance

6.2.1 Overview

A The SWIFT shall be inspected at the intervals described in this chapter.

B Depth maintenance of the SWIFT shall be the responsibility of Zelim and/or an endorsed maintenance organisation, and shall require the SWIFT to be removed from the vessel.

C When conducting 30-month depth maintenance inspections, follow the component inspection process in Section 6.2.5, noting any observations in Appendix B – 30-Month Depth Maintenance Record.

D This maintenance activity shall be carried out at a suitable facility.

E The objective of 30-month depth maintenance is to provide a full system inspection, incorporating checks, repairs, adjustments, or part replacements to maintain reliable operation and optimum performance of the safety equipment.

F Chapters 4–9 provide guidance on disassembly, cleaning, inspection and maintenance, testing, repair and replacement, and assembly procedures to enable full component assessment and replacement as required. Each component must be examined for both condition and performance.

G Once satisfied that the SWIFT is safe and operational, Zelim and/or the endorsed maintenance organisation shall issue a SWIFT Recovery System 30-Month Depth Maintenance Certificate.

6.2.2 SWIFT removal and transportation

6.2.2.1 Overview

1 The SWIFT shall be removed from the vessel and transported to a suitable location for 30-month depth maintenance. Refer to the SWIFT Survivor Recovery System Installation Guide (PN/SWI/010-00).

2 The maintenance facility shall complete a pre-disassembly inspection using Appendix B.1 (Pre-disassembly), recording the condition of the returned equipment and assessing whether any damage has occurred during transit.

6.2.3 30-month depth pre-maintenance checks

1 Complete the pre-disassembly process in accordance with Chapter 4.4 – Pre-disassembly and Appendix B.1 (Pre-disassembly), recording the condition of the returned equipment and assessing whether any damage has occurred during transit.

2 Any immediate concerns should be raised with Zelim as required and /or if the product warranty is in question.

3 The maintenance organisation shall ensure that all manuals and relevant documentation are maintained at the current revision level.

4 The maintenance organisation shall record all elements of maintenance, including the condition review, in the in the 30-month depth maintenance record (Appendix B).

NOTE The serial number of the SWIFT is located on the data plate at the end of the chassis, adjacent to the motor.

6.2.4 30-month depth maintenance process

1 Review all customer maintenance records and service paperwork to understand prior servicing and identify any reported faults since the last maintenance.

2 Review overall condition of the SWIFT and components, recording observations and issues.

3 Disassemble the SWIFT in accordance with Chapter 4 – Disassembly, removing all major sub-assemblies (foam moulds, chassis components, conveyor, winch, electrical enclosure, and control box).

4 Clean all components in accordance with Chapter 5 – Cleaning.

5 Conduct a detailed inspection of each component for:

A Corrosion, wear, or distortion of metallic components.

B Degradation, delamination, or water ingress in foam moulds.

C Condition of drive shafts, bearings, sprockets, and tension assemblies.

D Integrity of welds, fasteners, and chassis mounting points.

E Condition and routing of electrical wiring, connectors, and seals.

6 Replace all consumables and service items, including the following:

A Bearings, seals, and fasteners identified as worn or corroded.

B Lubricants and grease (lithium EP2) as required.

7 Inspect and test the winch assembly (if returned):

A Fully unspool the winch rope. Inspect the rope along its full length.

B Check the rope is securely retained at the winch rope clamp. Ensure the rope is retained by both clamping screws and torque the M8 × 12 rope clamp screws to 16 N·m (dry, no Loctite).

C Inspect the rope for fraying, furring, thinning, glassing (heat damage), dirt, or oil contamination.

D Check for zinc plating loss on the winch drum side cheeks where the rope makes contact.

E Ensure the winch is adequately lubricated. Apply lithium EP2 grease to the handle pivot, drum bearings, and any exposed gear or pawl interfaces as required.

CAUTION Avoid contaminating the winch drum or rope with grease.

F Operate the winch to pay out and recover the rope, confirming smooth engagement, no slipping, and positive ratchet/brake holding in both directions.

8 Reassemble the SWIFT in accordance with Chapter 9 – Assembly, applying specified torque values and Loctite 243 to all threaded fasteners.

9 IAW Chapter 7.1, complete a functional test.

10 Inspect the motor and gearbox for lubrication leaks or signs of oil contamination.

11 Complete the component inspection process in accordance with Chapter 6.2.5, then repackage the SWIFT in the shipping crate, taking internal and external photos of the SWIFT and shipping crate to record the condition prior to dispatch.

6.2.5 Component inspection process

The following process is to be used for inspecting the SWIFT and its components during 30-month depth maintenance.

NOTE If any operational issues were observed during the pre-maintenance functional checks and/or within the customer maintenance records, these may indicate distortion or misalignment of the chassis frame.

6.2.5.1 Hinge mounting bracket (if returned)

1 Examine for any wear, damage, or corrosion.

2 Check all nuts and bolts are secure.

3 Examine both bearings for free rotation and lubricate using lithium EP2 grease via the grease nipples.

4 If fitted, examine the mounting feet hinge pins and lubricate with lithium EP2 grease.

6.2.5.2 Floatation foam moulds and clamps

1 Examine for damage.

2 Check that the adhesive used to join the floatation foam moulds is intact.

3 Inspect reflective tapes for security and adhesion.

6.2.5.3 Conveyor belt

1. Examine for any wear, cracks, or damage.
2. Check that all white 5 mm diameter plastic rods are correctly located and secure.

6.2.5.4 Tension shaft

1. Examine for distortion, damage, and corrosion.
2. Examine each white toothed sprocket and spacer for positioning, wear, cracking, or damage, and ensure all grub screws are secure.

6.2.5.5 Motor and gearbox

1. Inspect for any damage or corrosion.
2. Check for oil leaks.
3. Examine electrical cables and sockets for any damage to outer casings.

6.2.5.6 Drive shaft assembly

1. Examine for any distortion, damage, or corrosion.
2. Examine each white toothed sprocket and spacer for correct positioning, wear, cracking, or damage, and ensure all grub screws are secure.

6.2.5.7 Bearing mounting brackets and drive shaft bearings

1. Examine for any wear, cracks, damage, or corrosion.

6.2.5.8 Stub pivot axles

1. Examine for any damage, distortion, or corrosion.

6.2.5.9 Tape switch

1. Check that the tape switch is securely fitted within its track.
2. Examine for any damage or cracking.
3. Examine electrical cables and sockets for any damage to their outer casings.

6.2.5.10 Chassis

1. Examine the chassis on a flat, level surface for any distortion, damage, or cracking.

   CAUTION Check for any weld deterioration.

2. Examine for any wear, damage, or corrosion. Pay particular attention to the mating surfaces where components are fitted or welded.
3. Examine each foam mounting stub for thread damage or corrosion. Remove any thread-locking adhesive from the inner threads.
4. Examine tensioner brackets for any damage or corrosion.
5. Examine hinge lugs for any damage or corrosion.
6. Examine lifting eyes for any damage or corrosion, and clean threads as required.
7. Check that all support bar countersunk-head cap screws are secure.
8. Examine tie bars for any damage or distortion. Remove any thread-locking adhesive from the inner threads.
9. Examine clamp plates for any damage or corrosion.

6.2.5.11 Nylon fairlead and temporary rope

1. Examine for any wear, cracks, or damage.
2. Ensure the temporary rope is correctly threaded through the fairlead and secured in place.

6.3 Unscheduled maintenance

6.3.1 Overview

A Unscheduled maintenance of the SWIFT shall be the responsibility of Zelim and/or an endorsed maintenance organisation and may be conducted in situ. The SWIFT may be removed from the vessel if required.

NOTE If removal from the vessel is required, refer to Section 6.2.2 – SWIFT removal and transportation.

B Any components found to be unserviceable shall be repaired or replaced in accordance with Chapters 4 to 9.

NOTE All electrical maintenance, repair, or component replacement shall be undertaken by a suitably qualified marine engineer.

C When conducting unscheduled maintenance, record all maintenance completed and note any observations in Appendix C – Unscheduled Maintenance Record.

6.3.2 Unscheduled maintenance process

1 Take clear photos of the SWIFT in situ and the surrounding installation area, and store them locally as part of the maintenance records.

2 Record all unscheduled maintenance completed and note any observations in Appendix C – Unscheduled Maintenance Record.

7 TESTING

This chapter outlines the requirements for functional testing of the SWIFT. All testing shall be carried out in accordance with the procedures detailed in this chapter.

7.1 Functional testing

7.1.1 Conveyor belt

A Conveyor belt testing

1. Operate the motor and gearbox in both forward and reverse directions using the 3-push-button control box (Fig 20) for 5 minutes each way, confirming smooth and uniform operation.

Fig 20 3-push-button control box

2. Check for uniform speed and smooth operation.
3. Check for any grating noise, changes in pitch, or excessive heat build-up.
4. Check the motor and gearbox for oil leaks.
5. Check for any tracking issues or asymmetry in belt tension.
6. Observe the conveyor belt engaging with the sprockets on the drive shaft and tension shaft, and check for squeaks, rubbing, or heat build-up at the shaft bearings.
7. With the conveyor belt running, press the middle stop button on the 3-push-button control box. Ensure the conveyor belt stops instantly.
8. Operate the forward and reverse buttons on the 3-push-button control box to check the conveyor belt can still operate. Press the middle stop button. Ensure the conveyor belt stops instantly.

7.1.2 Emergency stop button

A Emergency stop button testing

1. With the conveyor belt running, operate the red emergency stop button (Fig 21) and confirm the conveyor belt stops immediately.

Fig 21 Emergency stop button

2. Reset the emergency stop and confirm normal operation can be restored.

7.1.3 Tape switch

A Tape switch testing

1. With the conveyor belt running, carefully operate the tape switch and confirm the conveyor belt stops immediately.
2. Release the tape switch and confirm normal operation can be restored.

7.1.4 Current draw

A Current draw testing

1. Current draw shall be measured using a clamp meter on the supply cable during a trial MOB recovery to identify abnormal electrical load indicative of mechanical resistance or motor wear.

7.1.5 Electrical control box

A Electrical control box testing[2]

CAUTION Isolate the SWIFT electrical supply at the vessel's distribution panel (for example, the SWIFT supply breaker or isolator) before any maintenance or cleaning activity. Confirm isolation before proceeding.

NOTE The electrical control box is isolated via the upstream vessel supply. The external red fuse indicator provides visual confirmation of fuse status but shall not be used as a primary means of isolation. Under normal circumstances there is no requirement to open the control box to isolate the system. Opening the electrical control box and any internal inspection or work shall be carried out by Zelim or an endorsed maintenance organisation only (Fig 22), and by a suitably qualified marine engineer.

1. Visually inspect the electrical control box enclosure for damage, corrosion, cracks, or compromised seals (Figs 22 and 23).

Fig 22 Outside of electrical control box

Fig 23 Inside of electrical control box

2. Confirm the enclosure is securely mounted and that all external fixings are present, intact, and secure.
3. Inspect all external cable glands for tightness, damage, or signs of water ingress.

2 This section applies to 30-month depth maintenance and unscheduled maintenance only. It does not apply to annual or monthly maintenance.

4. If any abnormal damage or evidence of moisture ingress is identified, repair or replace affected components in accordance with Chapter 8.1 – Repair.
5. Restore the vessel's electrical supply and confirm the external red fuse indicator is illuminated to indicate fuse continuity.
6. Check external red fuse indicator for visual confirmation of fuse status.
7. Confirm the enclosure remains closed, sealed, and locked during normal operation.
8. With the conveyor belt running, verify forward, reverse, and stop operation using the 3-push-button control box controls, confirming correct response and smooth operation.
9. With the conveyor belt running, operate the emergency stop button and confirm the conveyor belt stops instantly. Reset the emergency stop and confirm normal operation can be restored.
10. With the conveyor belt running, carefully operate the emergency tape switch and confirm the conveyor belt stops instantly. Release the tape switch and confirm normal operation can be restored.
11. If any abnormal operation is identified, isolate the supply via the vessel distribution panel and investigate the cause. Repair or replace affected components in accordance with Chapter 8.1 – Repair.
12. Restore the vessel's electrical supply and confirm the external red fuse indicator is illuminated to indicate fuse continuity.
13. Record the results and any observations in Appendix B – 30-month depth maintenance record or Appendix C – Unscheduled maintenance record, as applicable.

8 REPAIR AND REPLACEMENT

This chapter outlines the procedures for repairing or replacing SWIFT components found to be damaged, worn, or unserviceable. All work shall be performed in accordance with these instructions to maintain safety, reliability, and compliance.

8.1 Repair

1 Repairs shall only be carried out on the components listed below:

A Conveyor belt

B Electrical control box and components

C Manual winch

2 Any components found to be unserviceable shall be removed and replaced with serviceable items as listed above. No locally manufactured, modified, or substitute components shall be fitted in place of Zelim-approved parts.

3 If the chassis is found to be damaged, distorted, or suffering from excessive corrosion, the chassis shall be replaced.

8.1.1 Conveyor belt

A Conveyor belt repair

1 Lay the conveyor belt on a flat surface and remove the 5 mm white plastic rods from either side of the damaged area by gripping them with long-nose pliers and pulling them out just past the last damaged part (Fig 24 and Fig 25).

<u>**NOTE**</u> <u>If the conveyor belt is not already in two halves, refer to section 4.5.3.</u>

Fig 24 Identify single white plastic rod

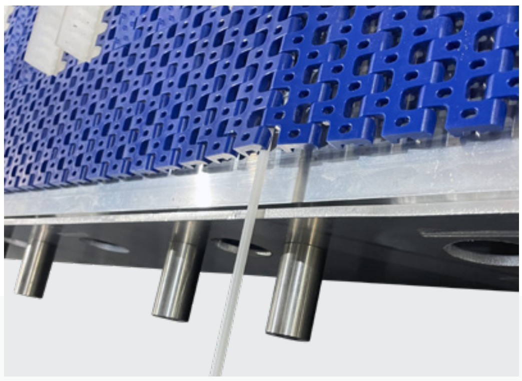

Fig 25 White plastic rod removal

2 Identify the like-for-like replacement parts (Fig 26). Remove the damaged section by carefully lifting it out with a flat-head screwdriver, then fit the new section in place, ensuring correct alignment with the adjoining parts.

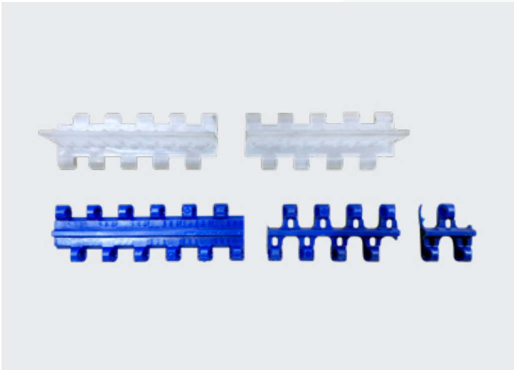

Fig 26 Conveyor belt spare parts

3 Slide both partially removed 5 mm white plastic rods back through the replaced section and the remaining interlocking parts until they reach the end of the belt.

8.1.2 Electrical control box

A 3-push-button control box repair

1 Any damage shall be repaired or replaced by a suitably qualified electrician.

B Emergency stop controller repair

1 Any damage shall be repaired or replaced by a suitably qualified electrician..

C Quick connectors repair

1 Any damage shall be repaired or replaced by a suitably qualified electrician.

D Motor and gearbox electrical cable repair

1 Any damage shall be repaired or replaced by a suitably qualified electrician.

8.1.3 Manual winch

A Manual winch repair

1 If the manual winch is damaged or unserviceable and repair is considered feasible, contact swift.support@zelim.com for guidance.

8.2 Replacement

1 Replacement of components shall be limited to the following:

- **A** Chassis
- **B** Motor and gearbox
- **C** Stub pivot axle
- **D** Foam mounting stubs
- **E** Floatation foam moulds
- **F** Clamp plates
- **G** Conveyor belt 5 mm plastic rods
- **H** Tension shaft, white teethed sprocket and spacers
- **I** Drive shaft assembly, white teethed sprocket and spacers
- **J** Tape switch
- **K** Lifting eyes
- **L** Securing pins
- **M** Hinge mounting bracket assembly
- **N** Vessel mounting bracket
- **O** Nylon fairlead and rope
- **P** Manual winch
- **Q** Manual winch rope
- **R** Ratchet spanner with 24 mm socket

NOTE Damaged components shall only be removed and replaced in accordance with this chapter and Chapter 4 – Disassembly.

NOTE The SWIFT may be required to be removed from the vessel and returned to Zelim and/or an endorsed maintenance organisation for replacement.

8.2.1 Chassis

A Chassis replacement

1 If the chassis (Fig 27) is found to be unserviceable, it cannot be repaired and shall be replaced with a new item.

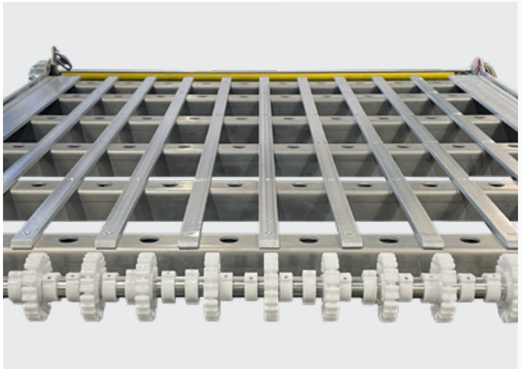

Fig 27 Chassis

8.2.2 Motor and gearbox

A Motor and gearbox removal

WARNING Prior to removing the motor and gearbox (Fig 27), ensure the motor is isolated from the power supply.

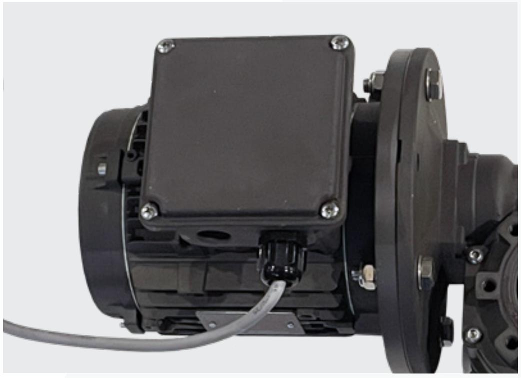

Fig 28 Motor and gearbox

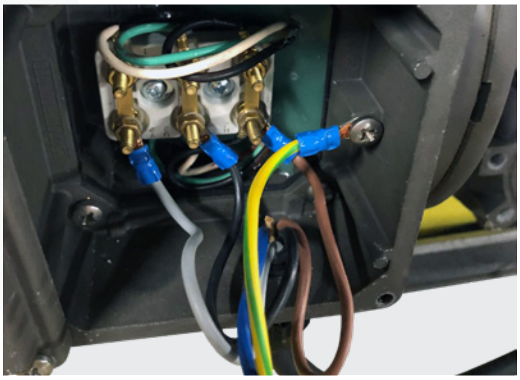

Fig 29 Power supply terminal box

1. Remove the four screws securing the cover on the power supply terminal box (Fig 28 and Fig 29).
2. Unscrew the cable gland nut, disconnect supply wires from the terminals and remove by gently pulling the cable through the gland.
3. Temporarily replace the gland nut and power supply terminal box cover.

NOTE Some motor and gearboxes have a quick disconnect socket so there is no requirement to disconnect the supply wires from the motor terminal box (Fig 30).

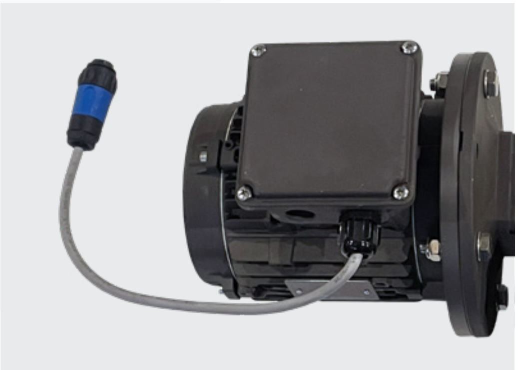

Fig 30 Quick disconnect socket

4 Remove the 4 × bolts securing the drive shaft cover from the gearbox (Fig 31).

Fig 31 Drive shaft cover

5 Prior to removing the motor and gearbox, remove the key bar securing the final drive shaft. The key bar prevents the formation of a partial vacuum between the close-fitting shaft and the final drive. To aid removal, temporarily screw an M6 threaded bolt into the key bar (Fig 32).

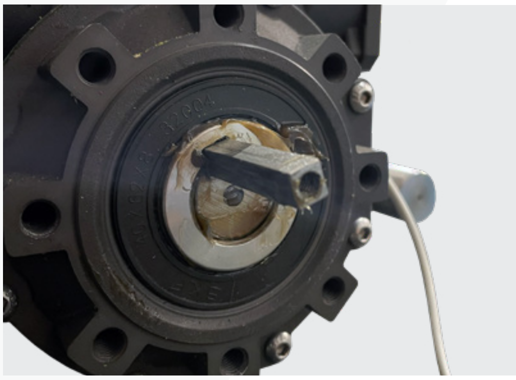

Fig 32 Key bar removal

6 The motor and gearbox are attached to the chassis via the bearing mounting brackets and drive shaft bearings.

7 To disconnect the motor and gearbox, remove the 8 × bearing mounting bracket Nyloc nuts (Fig 33).

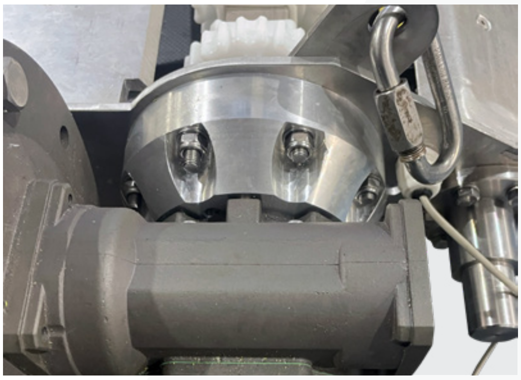

Fig 33 Isolated motor and gearbox

8 The lifting eye bracket/hinge lug is retained by the bearing mounting bracket bolts (Fig 34).

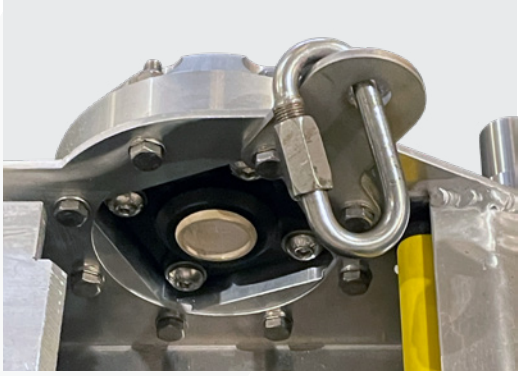

Fig 34 Lifting eye bolts

9 The motor and gearbox can now slide off the bearing drive bracket. If the fit is tight, use a soft plastic mallet to aid removal.

NOTE At this point the whole drive shaft may pull out of the chassis when attempting to remove the motor and gearbox. If this happens, continue to remove the motor and gearbox.

B Motor and gearbox replacement

1. Attach the bearing mounting bracket and drive shaft bearing to the motor and gearbox.
2. Locate the drive shaft assembly in the chassis and secure with 4 × M10 plain washers, 4 × M10 hexagonal bolt and replacement 4 × hexagonal M10 Nyloc nuts.
3. Line up the drive shaft gearbox final drive and drive shaft and insert key bar. Ensure key bar is coated with lithium EP 2 grease (Fig 32).
4. Fit the drive shaft cover to the gearbox, coating each M8 hexagonal bolt and plain washer with thread-locking adhesive (Loctite 243), and secure the cover (Fig 31).
5. Unscrew the cable gland nut, pass all cables through, and connect the supply wires to the motor terminals, if required.
6. Fit the cover to the power supply terminal box, coating each of the 4 × screws with thread-locking adhesive (Loctite 243), and secure the cover, if required (Fig 28).
7. Replace the gland nut and cover, if required.

8.2.3 Stub pivot axle

A Stub pivot axle removal

1. Remove the 6 × M8 hexagonal head bolts, 12 × M8 plain washers, and 6 × M8 hexagonal Nyloc nuts.
2. Lift off from the chassis (Fig 35).

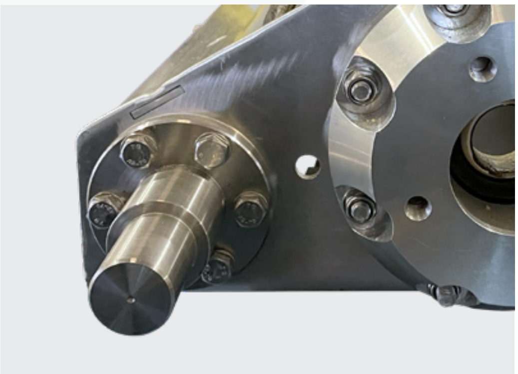

Fig 35 Stub pivot axle

B Stub pivot axle replacement

1. Attach replacement to the chassis and secure with 6 × M8 hexagonal head bolts, 12 × M8 plain washers and replacement 6× M8 hexagonal Nyloc nuts.

8.2.4 Foam mounting stubs

A Foam mounting stubs removal

1. Each chassis has 16 × standard size foam mounted stubs (Fig 36) and 2 × short foam mounted stubs across both sides of the chassis.

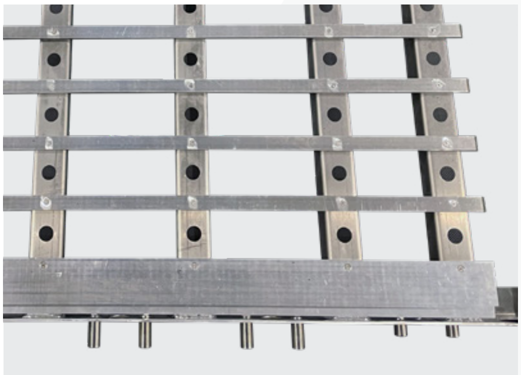

Fig 36 Foam mounting stubs

2. The 2 × short foam mounted stubs are located directly behind the motor body.
3. To replace a foam mounted stub, undo the Nyloc nut and washer, access is from inside the chassis.
4. Some foam mounting stubs may require gripping to loosen the fixing. If required, ensure the surface of the stub is protected from tooling, marks.

B Foam mounting stubs replacement

1. Apply thread locking adhesive (Loctite 243) to each foam mounting stub and pass through the chassis.
2. Attach the foam mounting stubs to the chassis with 1 × M12 plain washer and replacement M12 hexagonal Nyloc nut.

8.2.5 Floatation foam moulds

A Floatation foam mould removal

1. To remove a flotation foam mould, refer to Section 4.5.2 – Flotation foam moulds removal.

B Floatation foam moulds replacement

1. To replace a flotation foam mould, refer to Section 9.1.3 – Floatation foam moulds assembly.

8.2.6 Clamp plates

A Clamp plates removal

There are 10 clamp plates: six of type 1, two of type 2, and two of type 3 (Fig 37).

Fig 37 Clamp plates

1 Remove the M12 hexagonal bolts and plain washers from the chassis foam mounted stubs and lift away (Fig 38).

Fig 38 Foam clamp plate removed

B Foam clamp plate replacement

1 Fit the replacement foam clamp plate to the foam mould and secure it to the chassis foam mounting stub using M12 plain washers and M12 hexagonal bolts. Apply thread-locking adhesive (Loctite 243) and torque each bolt to 70 N·m.

8.2.7 Conveyor belt 5 mm plastic rods

A Conveyor belt 5 mm plastic rods removal

1 To replace a conveyor belt plastic rod, lift the edge of the conveyor belt to reveal the underside, identify a single white 5 mm diameter plastic rod connecting the modular system together, remove the white plastic rod by gripping it with long-nose pliers and pulling it out (Fig 39).

Fig 39 Identify single white plastic rod

B Conveyor belt plastic rods replacement

1 To replace a white plastic rod, temporarily secure the conveyor belt together with cable ties (Fig 40).

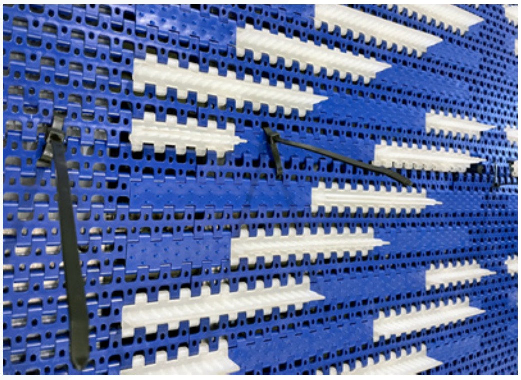

Fig 40 Conveyor belt temporary cable ties

2 Lift the edge of the belt and feed the white plastic rod through all sections of the conveyor belt until it becomes one piece (Fig 41).

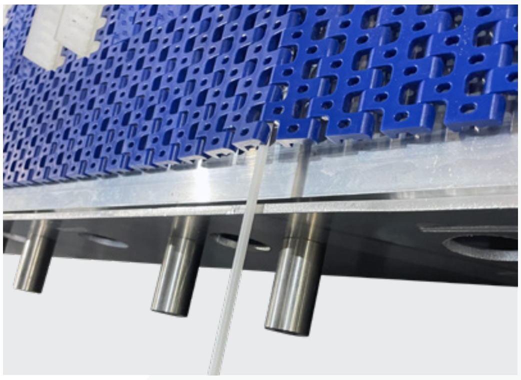

Fig 41 Single white plastic rod

3 Remove all temporary cable ties.

8.2.8 Tension shaft assembly

A Tension shaft assembly removal

1 With the conveyor belt removed from the chassis (Fig 42), release the 4 × M8 hexagonal Nyloc nuts and plain washers (Fig 43) securing the tensioner bracket and tension shaft bearing (Fig 44).

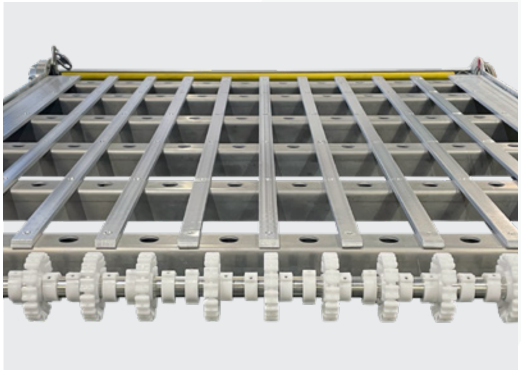

Fig 42 Chassis

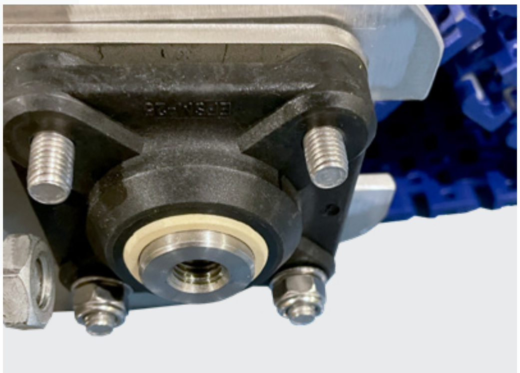

Fig 43 Nyloc nut removal

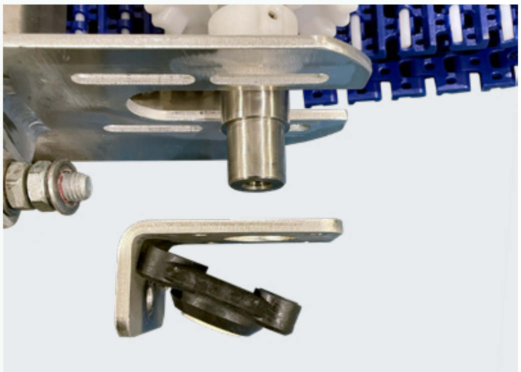

Fig 44 Tensioner bracket and tension shaft bearing

2 With the tension bracket and tension shaft bearing removed, the tension shaft can now be removed from the chassis.

WARNING This is a two-person lift due to the weight of the tension shaft.

3 To remove white teethed sprockets and spacers from the tensions shaft (Fig 45), undo small screws from the spacers (Fig 46) and slide off.

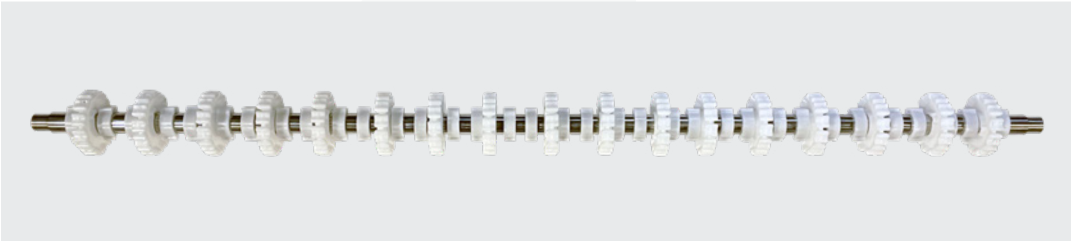

Fig 45 Tension shaft

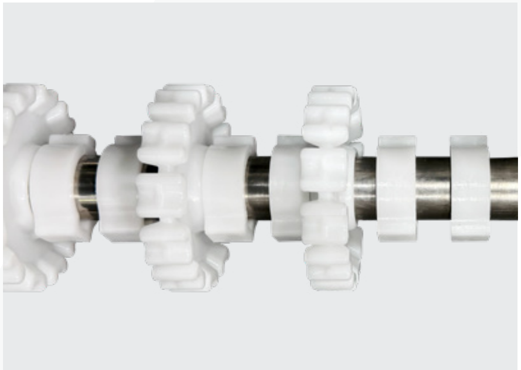

Fig 46 White teethed sprockets and spacers

B Tension shaft replacement

1 Fit the white toothed sprockets to the tension shaft, spacing them equally, and secure them in place with M5 hexagonal socket grub screws.

2 Carefully fit the tension shaft into the chassis frame.

WARNING This is a two-person lift due to the weight of the tension shaft.

3 Refit the tension bracket and tension shaft bearing over the tension shaft, and fit 4 × replacement M8 hexagonal Nyloc nuts and 4 × plain washers.

8.2.9 Drive shaft assembly

A Drive shaft assembly removal

1. Remove the motor and gearbox in accordance with Section 8.2.2 A – Motor and gearbox removal.
2. Remove Nyloc nuts, plain washers, and M10 hexagonal socket head cap screws from the bearing mounting bracket and hinge lug (Fig 47).

NOTE The use of a 13 mm S spanner will be required for the removal of the Nyloc nuts.

Fig 47 Bearing mounting bracket

3. The drive shaft can now be removed from the chassis (Fig 48).

WARNING This is a two-person lift due to the weight of the drive shaft.

Fig 48 Drive shaft

4. To remove white teethed sprockets and spacers, undo small screw and remove from the drive shaft.

B Drive shaft replacement

1 Fit the white toothed sprockets to the drive shaft, spacing them equally, and secure them in place with M5 hexagonal socket grub screws (Fig 49).

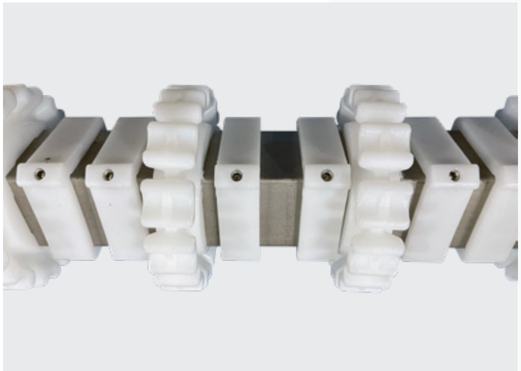

Fig 49 White teethed sprockets and spacers

2 Refit the bearing mounting bracket and hinge lug to the chassis, and secure using plain washers, M10 hexagonal socket head cap screws, and new M8 hexagonal Nyloc nuts.

3 Carefully fit the drive shaft into the worm drive mounting bracket and bearing.

WARNING This is a two-person lift due to the weight of the drive shaft.

4 Refit the motor and gearbox in accordance with Section 8.2.2 B – Motor and gearbox replacement.

8.2.10 Tape switch

A Tape switch removal

1 Unfasten the gland nut from the chassis, adjacent to the gearbox (Fig 50).

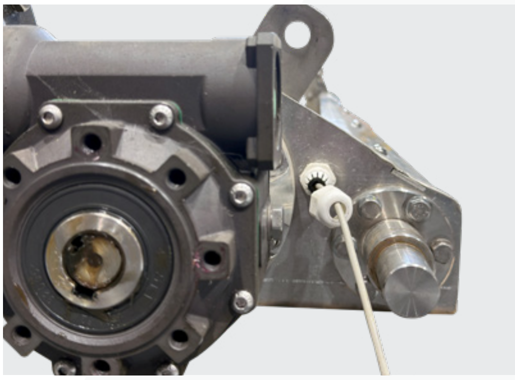

Fig 50 Gland nut

2 Release the connection plug from the control box wiring harness (Fig 51), unscrew the plug cap and release the wires from the terminals, as required.

Fig 51 Inside of electrical control box

3 Later SWIFTs have a quick disconnect socket so there is no requirement to disconnect the supply wires from control box (Fig 52).

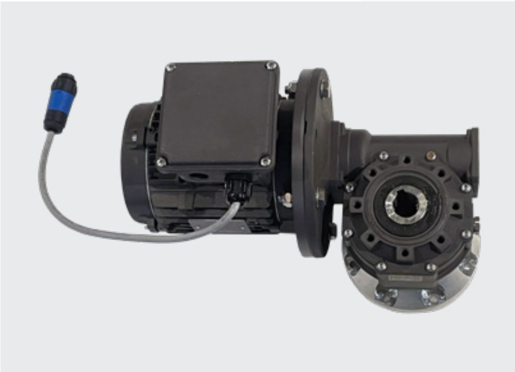

Fig 52 Quick disconnect socket

4 Starting at the end of the tape switch furthest away from the cable, gently peel out the tape switch from the mounting track (Fig 53).

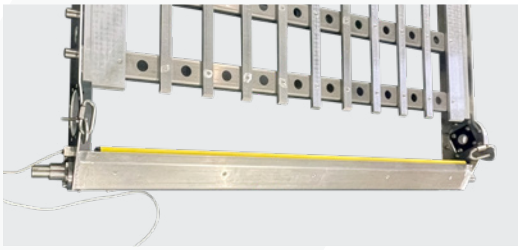

Fig 53 Tape switch

5 When fully released, pull the cable through the gland.

B Tape switch replacement

1 Starting at the end of the tape switch, furthest away from the cable, gently fit the tape switch into the mounting track.

2 Re-attach the connection plug to the control box wiring harness (if required) and secure the wires to the terminals as shown in Appendix D – Control box wiring diagram.

NOTE Some tape switches incorporate a quick-disconnect socket which shall be re-attached during reassembly.

8.2.11 Lifting eye

A Lifting eye replacement

1 If a lifting eye is found to be unserviceable, it cannot be repaired and shall be replaced with a new item (Fig 54).

Fig 54 Lifting eye

8.2.12 Securing pins

A Securing pin replacement

1 If a securing pin is found to be unserviceable, it cannot be repaired and shall be replaced with a new item.

8.2.13 Hinge mounting bracket assembly

A Hinge mounting bracket replacement

1 If a hinge mounting bracket is found to be unserviceable, it cannot be repaired and shall be replaced with a new item (Fig 55).

Fig 55 Hinge mounting bracket

8.2.14 Vessel mounting bracket

A Vessel mounting bracket replacement

1 If a vessel mounting bracket (Fig 56) is found to be unserviceable, it cannot be repaired and shall be replaced with a new item.

Fig 56 Vessel mounting bracket

8.2.15 Nylon fairlead and rope

A Nylon fairlead and rope removal

1 If a nylon fairlead is found to be unserviceable, undo dome head cap screws and pull rope out (Fig 57).

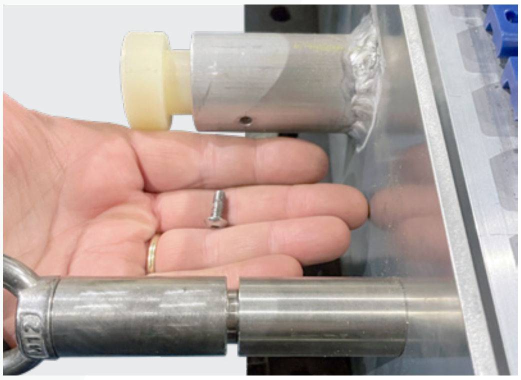

Fig 57 Nylon fairlead

2 Pull the rope out through the chassis.

B Nylon fairlead and rope replacement

1. Attach nylon fairlead to the chassis, apply thread locking adhesive (Loctite 243) to the dome head cap screws and secure.
2. Thread the nylon rope through each of the nylon fairleads and secure both ends with an overhand stop knot (Fig 58).

Fig 58 Overhand stop knot

8.2.16 Manual winch

A Manual winch replacement

1 If the manual winch is found unserviceable and beyond repair (Fig 59), it must be replaced.

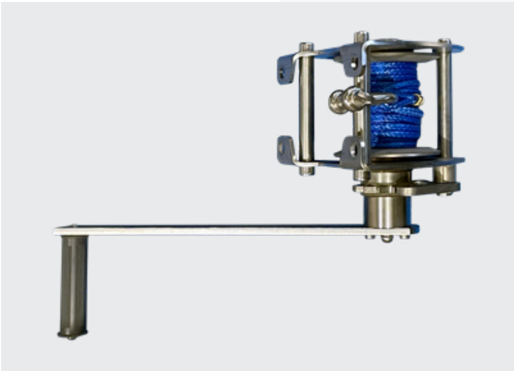

Fig 59 SWK-LB-250-4-20M manual winch

8.2.17 Manual winch rope

A Manual winch rope removal

1 Unspool all the Dyneema® rope from the winch drum.

2 Undo both M8×12 clamping screws (Fix 60) and remove rope from the winch.

Fig 60 M8×12 clamping screws (SWK-LB-650-6-20M manual winch)

B Manual winch rope replacement

1 Feed the replacement Dyneema® rope into the rope termination hole and secure it with both M8×12 clamping screws. Torque to 16 N·m (dry, no Loctite).

WARNING When attaching the Dyneema® rope (Fig 61), ensure it feeds beneath the winch drum and over the wheel. Incorrect routing will prevent correct operation.

Fig 61 Correctly routed Dyneema® rope

2 Wind the remaining rope onto the drum, ensuring it is evenly distributed across the drum surface.

CAUTION Ensure at least five full turns of rope remain on the drum when the SWIFT is in the lowest load position.

3 Check that the winch operates smoothly and uniformly when spooling in and out.

8.2.18 Ratchet spanner and 24 mm socket

A Ratchet spanner and 24 mm socket replacement

1 If the ratchet spanner and 24 mm socket are found to be unserviceable, they must be replaced with new items rather than repaired (Fig 62).

Fig 62 Ratchet spanner and 24 mm socket

8.3 Bolt and nut torque requirements

1 Table 4 below lists the required bolt and nut torque values for maintenance and installation activities.

CAUTION Loctite 243 must not be applied when tightening bolts fitted with Nyloc nuts.

The torque values listed cover all standard SWIFT fasteners and installation fixings, including those used for vessel mounting.

Table 4 Bolt torques and spanner sizes[3]

BOLT DIAMETER	MATERIAL	TIGHTENING TORQUE	SPANNER / SOCKET SIZE
M6	Stainless steel A4-70	8.5 N·m	10 mm
M8	Stainless steel A4-70	20 N·m	13 mm
M10	Stainless steel A4-70	40 N·m	17 mm
M12	Stainless steel A4-70	70 N·m	19 mm
M14	Stainless steel A4-70	115 N·m	22 mm
M16	Stainless steel A4-70	180 N·m	24 mm
M20	Stainless steel A4-70	370 N·m	30 mm

3 All torque values apply to lubricated threads assembled with Loctite 243 unless otherwise stated. When using Nyloc nuts, Loctite 243 must not be applied; torque values may be used as a guide, but actual preload will be lower due to increased friction. If tightening dry (no lubricant or threadlocker), increase torque by approximately 10–15 % to achieve equivalent preload.

9 ASSEMBLY

This chapter details the procedures required to assemble the SWIFT following maintenance or component replacement. All assembly operations shall be performed in accordance with the instructions provided in this chapter to ensure correct function and safety of the equipment.

NOTE Any Nyloc nuts that have been loosened or removed during maintenance shall be discarded and replaced with new ones.

9.1 General

9.1.1 Nylon fairlead

A Nylon fairlead assembly

1. Thread the nylon rope through the chassis.
2. Attach each nylon fairlead to the chassis, apply thread-locking adhesive (Loctite 243) to the dome head cap screws, and secure them (Fig 63).

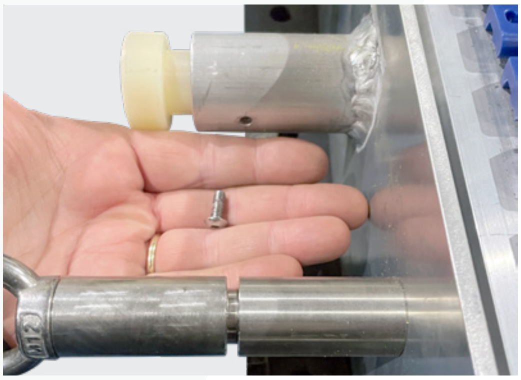

Fig 63 Nylon fairlead

3 Thread the nylon rope through each of the nylon fairleads and secure both ends with a overhand stop knot (Fig 64).

Fig 64 Overhand stop knot

9.1.2 Conveyor belt

A Conveyor belt assembly

1 Place the chassis on a flat surface (Fig 65).

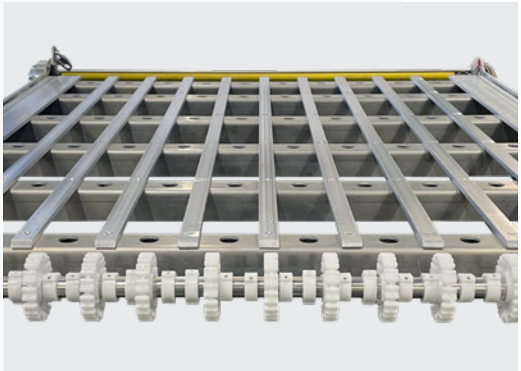

Fig 65 Chassis

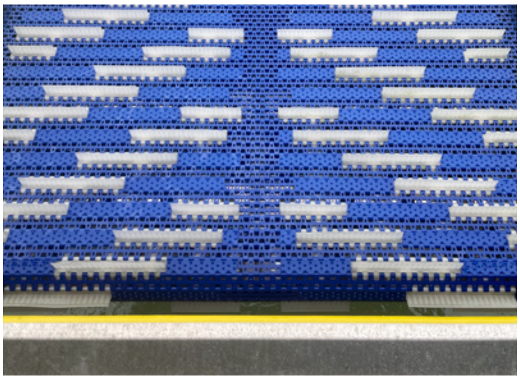

Fig 66 Correctly positioned conveyor belt on chassis

2 With assistance from a second person, slowly feed the conveyor belt under the chassis.

NOTE Ensure the conveyor belt is facing the correct direction and the correct way up, with the white fins uppermost in a V-shape arrow, pointing away from the tape switch (Fig 66).

3 Position the conveyor belt over the white plastic drive shaft sprockets and align both ends of the belt, ensuring the conveyor belt is centred over the drive and tensioner shaft sprockets.

4 Connect a 24 mm socket and ratchet to the drive and crank clockwise, feeding approximately 300 mm of belt onto the upper chassis.

5 The 5 mm white nylon rod running along the underside of the belt shall sit correctly in the grooves of the sprockets on both drive shaft and tension shaft.

CAUTION If you proceed to tension before the 5mm white nylon rod is correctly seated in the white sprockets, it will cause belt distortion (Figs 67 and 68) and will require the tension to be released and the belt reset.

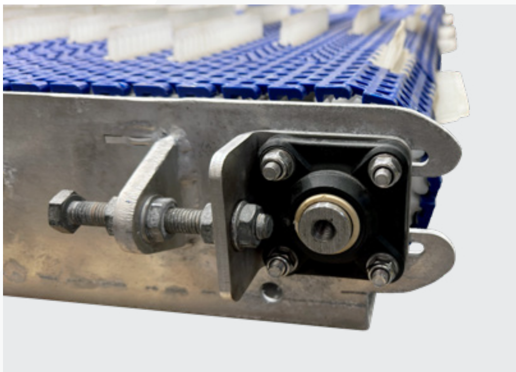

Fig 67 Belt distortion seen next to white fins on outer curve

Fig 68 Close up of belt distortion

6 Pull in both ends and temporarily secure the conveyor belt with a number of cable ties (Figs 69 and 70) to allow the 5 mm diameter plastic rod to pass through.

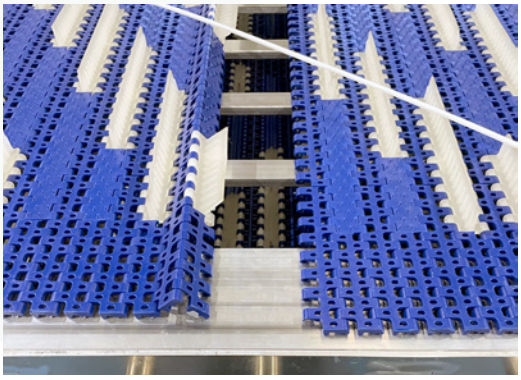

Fig 69 Conveyor belt alignment prior to cable tie installation

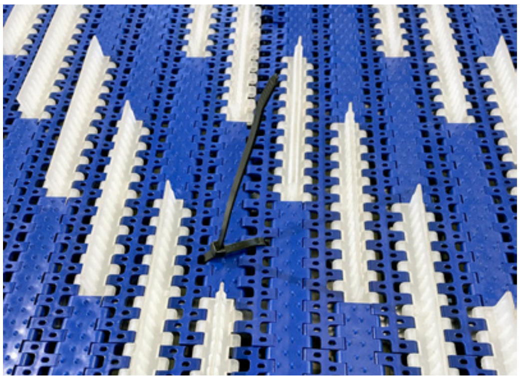

Fig 70 Cable tie temporary securing conveyor belt

7 Lift the edge of the belt and insert a new 5 mm diameter plastic rod through all belt sections until it forms a single continuous loop.

NOTE Check that both white nylon rods are still correctly seated in the grooves of the sprockets on both the drive shaft and tension shaft.

8 Remove all temporary cable ties.

9 Check and confirm that there is equal clearance between the conveyor belt edges and the chassis (approximately 27 mm on each side).

10 Using a 24 mm socket and ratchet, manually rotate the conveyor belt via the M16 drive nut on the drive shaft to confirm that the belt is moving freely.

11 The sprockets feature a raised profile that engages with a corresponding groove on the underside of the conveyor belt to maintain correct lateral tracking.

NOTE The spacing between the clamps allows the drive shaft sprockets to self-align with the conveyor belt.

12 Adjust the conveyor belt tension evenly via both tensioner brackets until 10 threads are visible between the chassis and the tensioner bracket (Fig 71). Ensure that a minimum of one and a maximum of three full threads are visible beyond each nut after tightening.

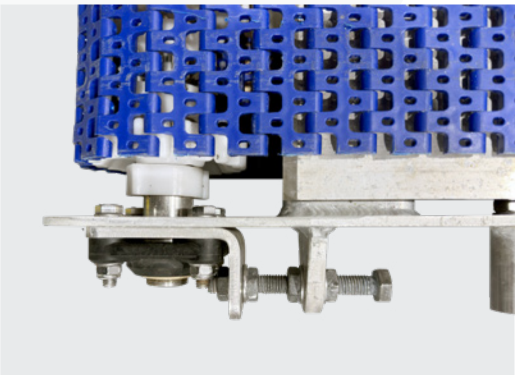

Fig 71 10 threads visible

13 Tighten 1 × M12 hexagonal bolt, 4 × M12 hexagonal half nuts, and 4 × M12 plain washers on each side.

14 Using a 13 mm spanner and a torque wrench fitted with a 13 mm socket, tighten the tension shaft bearing using 4 × M8 Nyloc nuts to a torque of 28 N·m.

9.1.3 Floatation foam

A Floatation foam moulds assembly

1. Refit the 2 × lifting eyes.
2. Using a mechanical winch, position the chassis with the motor and gearbox located at the bottom (Fig 72).

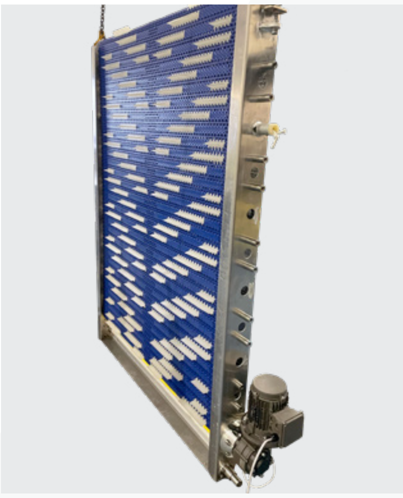

Fig 72 Chassis position

3. Carefully position and fit foam mould No. 2 over the motor and gearbox (Fig 73), then fit foam mould No. 3 (Fig 74) on the opposite side.

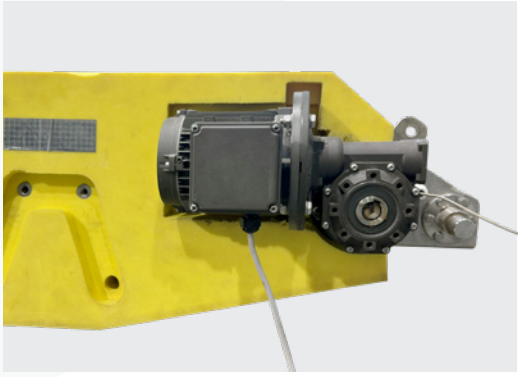

Fig 73 Foam mould No. 2

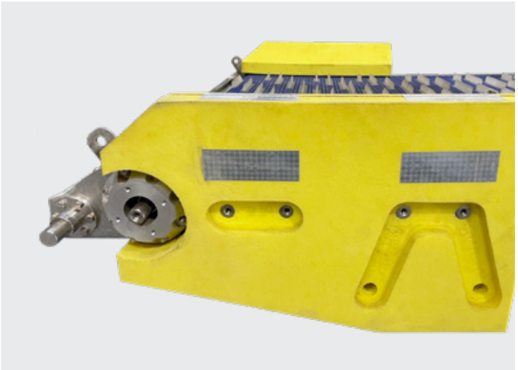

Fig 74 Foam mould No. 3

4 Insert 2 × tie bars through foam mould No. 2 and into foam mould No. 3, aligning them as shown in Fig 75.

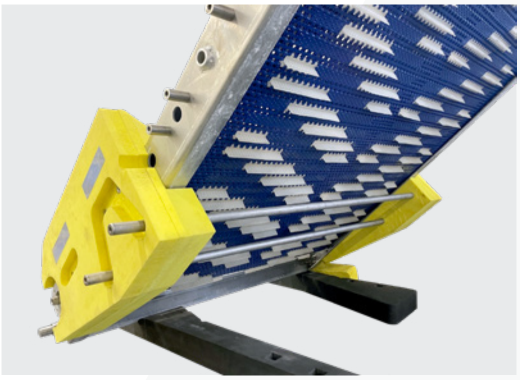

Fig 75 Tie bars

5 Using a torque wrench fitted with a 19 mm socket, secure clamp plates Nos. 1 and 3 to both sides of the chassis foam-mounting stubs and tie bars using M12 plain washers and M12 hexagonal bolts. Apply thread-locking adhesive (Loctite 243) and tighten each bolt to a torque of 70 N·m.

6 Fit 4 × foam moulds No. 1 to the chassis.

7 Locate 12 × foam moulds No. 6.

8 Partially insert a tie bar into the lower position of either foam mould No. 1, then fit the 12 × black foam moulds No. 6. Locate the opposite-side foam mould No. 1.

NOTE Ensure they are fitted in the correct orientation, with the edge having the smaller gap (40 mm) from the holes positioned closest to the conveyor belt (Fig 76).

Fig 76 Foam mould

9 Insert a further 3 × tie bars through foam moulds No. 1 and No. 6 and through the chassis, aligning them with the opposite-side foam mould No. 1.

10 Using a torque wrench fitted with a 19 mm socket, secure clamp plates Nos. 1 to both sides of the chassis foam-mounting stubs and tie bars using M12 plain washers and M12 hexagonal bolts. Apply thread-locking adhesive (Loctite 243) and tighten each bolt to a torque of 70 N·m.

11 Lower the chassis onto a flat surface using the mechanical winch. Use sacrificial pieces of foam to protect the foam moulds from chains or slings as required.

12 Fit foam mould No. 5 (non-motor side) to the chassis, and insert the 12 × black/yellow foam moulds No. 8 together with the tie bars. Fit foam mould No. 4 (motor side) to the chassis.

13 Fit the remaining 2 × tie bars through foam moulds No. 4 and No. 5.

14 Using a torque wrench fitted with a 19 mm socket, secure clamp plates Nos. 2 and 4 to the chassis foam-mounting stubs and tie bars using M12 plain washers and M12 hexagonal bolts. Apply thread-locking adhesive (Loctite 243) and tighten each bolt to a torque of 70 N·m.

9.1.5 Hinge mounting

A Hinge bracket

1 If the hinge mounting bracket was returned with the SWIFT and removed during disassembly, it must now be refitted.

2 Position the hinge mounting bracket on the chassis and align it with the hinge bearings and stub axles. Insert the 4 × M16 stainless steel bolts, fit the corresponding plain washers and nuts, and finger-tighten loosely.

9.2 Post assembly checks

9.2.1 Tension check

A Post assembly tension check

1. Conduct a general static tension check by gripping the conveyor fins halfway along the conveyor belt and pulling at 90° to the SWIFT longitudinal axis.
2. Note the distance and force required, and compare these with the pre-maintenance conveyor belt tension.

9.3 Documentation

A Post assembly documentation

1. Record all steps and findings of the maintenance process, including the condition review, in the 30-Month Depth Maintenance Record (Appendix B).
2. Record any parts replaced and clearly state any advisory notes.
3. Raise any product warranty concerns with Zelim at swift.support@zelim.com.

10 STORAGE AND DISPOSAL

This chapter outlines the correct procedures for storing and disposing of the SWIFT and its components. Proper storage protects the equipment from environmental damage and contamination, while disposal must be carried out responsibly and in accordance with local regulations.

10.1 Storage

1. If the SWIFT is to be removed from the vessel for an extended period, it shall be cleaned in accordance with Chapter 5 – Cleaning.
2. The SWIFT shall be stored in a dry, clean area, away from acids, oils, and greases.

10.2 Disposal

1. SWIFT metal and plastic components shall be sent for recycling where possible.
2. All other components shall be disposed of IAW local procedures.

11 TOOLS, EQUIPMENT AND MATERIALS

This chapter lists the tools, equipment, and materials required for the maintenance, repair, assembly, and installation of the SWIFT. Only the items specified in this chapter are approved for use to ensure safety, reliability, and compliance with maintenance standards.

11.1 Tools

1 Tools required for SWIFT maintenance and installation are listed in Table 5.

Table 5 Tools required for maintenance and reassembly.

REF	ITEM	NOTES / SPECIFICATION
A	Spanner 8 mm (M5)	
B	Spanner 10 mm (M6)	
C	Spanner 13 mm (M8)	
D	Spanner 13 mm (M8, S-shape)	For restricted-access areas.
E	Spanner 17 mm (M10)	
F	Spanner 19 mm (M12)	
G	Spanner 24 mm (M16)	
H	Spanner 30 mm (M20)	For hinge-mounting bracket and deck-mount installation bolts
I	Torque wrench (0 – 100 N·m)	For M6–M10 fasteners
J	Torque wrench (40 – 200 N·m)	For M12–M16 maintenance fasteners
K	Torque wrench (100 – 400 N·m)	For M20 installation bolts only
L	Allen key 4 mm	
M	Long-nose pliers	For gripping and wire access
N	Locking-grip pliers	For holding or freeing seized components
O	Tape measure	For dimensional checks and alignment
P	Soft-face mallet	For seating bearings and components without damage
Q	Grease gun	For applying lithium EP2 grease
R	Clamp meter	For measuring supply current

REF	ITEM	NOTES / SPECIFICATION
S	Screwdriver (flat-blade)	For control box screws
T	Screwdriver (Phillips)	For control box and motor and gearbox terminal covers
U	Multimeter	For electrical continuity and control-circuit checks

11.2 Equipment and materials

1 Equipment and materials required for SWIFT maintenance, cleaning, and reassembly are listed in Table 6.

Table 6 Approved equipment and materials for maintenance and reassembly.

REF	ITEM	NOTES / SPECIFICATION
A	Thread-locking adhesive — Loctite 243 (Blue)	For stainless-steel fasteners; not to be used with Nyloc nuts
B	Lithium EP2 grease (marine grade)	For general lubrication of bearings, shafts, winch, gearbox, and moving parts
C	Cable ties	For securing wiring and the conveyor belt
D	Nylon fairlead rope	Temporary handling line for maintenance operations
E	Lifting slings (rated ≥ 1000 kg)	For controlled lifting of the SWIFT assembly
F	8 mm Dyneema® winch rope	For manual winch SWK-LB-250-4-20M
G	8 mm Dyneema® winch rope	For manual winch SWK-LB-650-6-20M
H	Soft cleaning brushes	For removing debris and corrosion during cleaning
I	Non-abrasive cloths	For drying and surface finishing after washdown
J	Mild detergent suitable for a marine environment	For cleaning in accordance with Chapter 6 – Cleaning
K	Fresh water supply (hose or bucket)	For rinsing during post-cleaning
L	Adult MOB manikin	For functional testing and recovery simulation

12 COMPONENT PARTS AND MATERIALS

This chapter lists the component parts, assemblies, and materials used in the manufacture and maintenance of the SWIFT. Each item is identified by part number, quantity, and description to support correct replacement, ordering, and traceability.

12.1 SWIFT assembly parts list

Table 7 SWIFT assembly part numbers and descriptions.

ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	SWI-000	Chassis assembly
2	4	SWI-025-000	Foam mould 1
3	2	SWI-033-000	Tensioner bracket (stainless steel 316)
4	2	SWI-037-000	Tension shaft bearing
5	56	M12 × 1.75 × 25 mm	Hexagonal head bolt (stainless steel A4-70)
6	1	SWI-020-000	Worm drive mounting bracket
7	2	SWI-019-000	Drive shaft bearing
8	1	SWI-022-000	Drive gearbox
9	8	M8 × 1.25 × 25 mm	Hexagonal socket head cap screw (stainless steel A4-70)
10	64	M8 (Form A)	Plain washer (stainless steel A4)
11	8	M10 × 1.5 × 35 mm	Hexagonal socket head cap screw (stainless steel A4-70)
12	16	M10 (Form A)	Plain washer (stainless steel A4)
13	36	M8 × 1.25 mm	Hexagonal Nyloc nut (stainless steel A4)
14	60	M5 (Form A)	Plain washer (stainless steel A4)
15	60	M5 × 0.8 mm	Hexagonal Nyloc nut (stainless steel A4?)
16	55	M5 × 0.8 × 25 mm	Hexagonal socket flat head countersunk head cap screw (stainless steel A4-70)
17	1	SWI-021-000	Bearing mounting bracket
18	2	M12 x 1.75 x 80 mm	Hexagonal head bolt (galvanised steel grade 8.8)

ITEM	QTY	PART NUMBER	DESCRIPTION
19	1	SWI-023-000	Drive motor
20	1	SWI-032-000	Tensioner shaft assembly
21	1	SWI-016-000	Drive shaft assembly
22	11	SWI-012-000	SWIFT-2 support bar
23	2	SWI-015-000	Pivot axle
24	8	SWI-038-000	Tie bar
25	12	M8 × 1.25 × 35 mm	Hexagonal head bolt (stainless steel A4-80)
26	2	SWI-036-000	Clamp plate 3
27	82	M12 (Form A)	Plain washer (stainless steel A4)
28	6	M12 x 1.75 mm	Hexagonal half-nut (galvanised steel, grade 8.8)
29	12	SWI-031-000	Foam mould 8
30	1	SWI-040-000	Belt 3.6 m × S.25-801 SL PE, blue, 1340 mm wide
31	1	SWI-013-000	Support bar 2
32	1	SWI-014-000	Support bar 3
33	8	M8 × 1.25 × 40 mm	Hexagonal head bolt (stainless steel A4-70)
34	16	M8 × 1.25 × 50 mm	Hexagonal head bolt (stainless steel A4-70)
35	2	SWI-041-000	Hinge lug
36	1	SWI-029-000	Foam mould 5
37	1	SWI-024-000	Tape switch
38	6	SWI-034-000	Clamp plate 1
39	1	SWI-026-000	Foam mould 2
40	12	SWI-030-000	Foam mould 6
41	1	SWI-028-000	Foam mould 4
42	1	SWI-035-000	Clamp plate 2
43	1	SWI-043-000	Tape switch mounting angle
44	18	SWI-044-000	Chassis foam mounting stub
45	4	M10 × 1.5 mm	Hexagonal Nyloc nut (stainless steel A4)
46	4	M10 × 1.5 × 40 mm	Hexagonal Nyloc nut (stainless steel A4)

ITEM	QTY	PART NUMBER	DESCRIPTION
47	20	M12 × 1.75 mm	Hexagonal Nyloc nut (stainless steel A4)
48	2	SWI-045-000	Chassis foam mounting stub short
49	1	SWI-046-000	Clamp plate 4
50	1	SWI-027-000	Foam mould 3
51	2	SWI-042-000	Nylon fairlead
52	5	M5 × 0.8 × 16 mm	Hexagonal socket flat head countersunk screw (stainless steel A4-70)
53	2	M5 × TBC × TBC mm	Hexagonal socket flat head countersunk screw (stainless steel A4)
54	2	SWI-049-000	Open rubber grommet
55	1	SWI-048-000	Drive shaft
56	1	M16 × 2.0 × 48 mm	Hexagonal stud connector nut (stainless steel A4)
57	17	SWI-017-000	Sprocket, 12T, 100 mm, 40 × 40 mm, S.25, 800 N
58	34	SWI-018-000	40 × 40 mm nutural POM retainer ring
59	34	M5 × 0.8 × 4 mm	Hexagonal socket grub screw with flat point (stainless steel A4-70)
60	1	SWI-047-000	Tensioner shaft
61	11	SWI-039-000	Tensioner shaft cog
62	22	SWI-049-000	Retainer ring, Ø30 mm, POM (natural)
63	22	M5 × 0.8 × 4 mm	Hexagonal socket grub screw with flat point (stainless steel A4-70)
64	2	SWI-052-000	M12 lifting eye bolt
65	1	SWK-LB-250-4-20M	Hand winch (obsolete)
66	1	SWK-LB-650-6-20M	Hand winch
67	1	TBC	Winch rope (Dyneema®, 8 mm)

APPENDICES

A ANNUAL MAINTENANCE RECORD

Annual maintenance shall be carried out by Zelim and/or an approved maintenance organisation.

This record is provided for guidance only. The SWIFT Survivor Recovery System Maintenance Manual (PN/SWI/011-00, Version 2.0) shall be used when carrying out annual maintenance.

COMPANY		LOCATION	

MODEL		SERIAL		DATE	

A.1 Annual maintenance

Chapters 5 to 9 shall be used when carrying out annual maintenance activities.

NO	ITEM	OBSERVATIONS
1	**Photograph SWIFT** Take clear photos of the SWIFT in situ and the surrounding installation area.	
2	**Check monthly inspection checklists** Confirm all monthly inspection records are present, complete, and signed.	
3	**Inspect SWIFT and accessories** Inspect the SWIFT and accessories for corrosion, debris, or marine growth.	
4	**Inspect and lubricate mounting feet hinge pins** Remove, inspect, and lubricate mounting feet hinge pins with lithium EP2 grease.	

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NO	ITEM	OBSERVATIONS
5	**Lubricate mounting bracket bearings** Using a grease gun with lithium EP2 grease, lubricate the mounting bracket bearings via the grease nipples.	
6	**Check fastener torque** Confirm all accessible structural and mechanical fasteners are torqued to the specified values.	
7	**Check conveyor belt sprocket alignment** Check the drive and tension shaft sprockets are aligned with the conveyor belt.	
8	**Verify conveyor belt tracking/tension** Verify the belt tracks centrally on the sprockets while running, with no deflection, flutter, or vibration.	
9	**Check conveyor belt operation** Confirm smooth conveyor belt operation with no squeaks, rubbing, or heat build-up at the shaft bearings.	
10	**Check conveyor belt speed** Confirm the conveyor belt speed is consistent in forward and reverse.	
11	**Conduct functional test** Run motor and gearbox forward/reverse for 5 minutes each way. Note uniform speed, smoothness, and any abnormal noise, vibration, or heat build-up.	
12	**Check Emergency Stop button** Test the Emergency Stop button to confirm it isolates power correctly.	

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NO	ITEM	OBSERVATIONS
13	**Check 3-push-button control box** Test the 3-push-button control box (Forward, Reverse, Stop) and confirm Stop halts operation.	
14	**Check tape switch operation** Confirm the tape switch activates correctly to stop the SWIFT.	
15	**Inspect motor and gearbox** Inspect the motor and gearbox for lubrication leaks or signs of oil contamination.	
16	**Check SWIFT deployment** Deploy the SWIFT and observe for smooth, controlled operation.	
17	**Unspool winch rope** With SWIFT supported by buoyancy, fully unspool the winch rope.	
18	**Check winch rope retention** Ensure the winch rope is retained by both clamping screws and connected to the vessel strong point.	
19	**Inspect winch rope** Inspect for fraying, thinning, heat damage, dirt, or oil contamination.	
20	**Inspect winch drum plating** Inspect for zinc plating loss on the winch drum side cheeks where the rope contacts.	

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NO	ITEM	OBSERVATIONS
21	**Inspect nylon fairleads** Inspect nylon fairleads for wear where the rope passes through the chassis.	
22	**Grease winch** Apply lithium EP2 grease to the handle, bearings, and gears, avoiding contamination of the rope.	
23	**Check winch operation** Operate the winch and confirm smooth operation and positive holding.	
24	**Inspect control box** Isolate power and inspect control box for water ingress, corrosion, or loose connections.	
25	**Check electrical system** Inspect cables and terminations, then confirm correct operation after restoring power.	
26	**Conduct MOB trial** Conduct trial recovery of adult MOB manikin and record current draw.	

FURTHER COMMENTS

ENGINEER NAME	
ENGINEER EMAIL	

SIGNED

PAGE 5 OF 5

B 30-MONTH DEPTH MAINTENANCE RECORD

30-month depth maintenance, including the associated pre-disassembly inspection, shall be carried out by Zelim and/or an approved maintenance organisation.

The SWIFT Survivor Recovery System Maintenance Manual (PN/SWI/011-00) and Installation Guide (PN/SWI/010-00) shall be used when carrying out pre-disassembly inspection, depth maintenance, and reinstallation on the vessel.

B.1 Pre-disassembly inspection

Chapter 4.4 – Pre-disassembly shall be used when carrying out the inspection.

NO	ITEM	EQUIPMENT RETURNED?		CONDITION		
				GOOD	FAIR	POOR
1	Shipping crate	Y	N			
2	SWIFT	Y	N			
3	Vessel hinge mounting bracket	Y	N			
4	Winch (including rope)	Y	N			
5	Control box	Y	N			
6	3-push-button control box	Y	N			
7	Emergency stop button	Y	N			
8	Electrical cabling	Y	N			
9	Securing pins (×2)	Y	N			
10	Ratchet spanner and 24 mm socket	Y	N			

B.2 30-month depth maintenance

Chapters 4 to 9 shall be used when carrying out 30-month depth maintenance activities.

NO	ITEM	OBSERVATIONS
1	**Photograph SWIFT** Take clear photos of the SWIFT (including data plate showing serial number) and any returned components.	
2	**Pre-disassembly functional test** Complete a pre-disassembly functional test and confirm baseline operation.	
3	**Clean the SWIFT assembly** Clean the SWIFT assembly thoroughly prior to maintenance.	
4	**Check service history** Review all customer maintenance records and service paperwork.	
5	**Condition review** Review overall condition of SWIFT and components. Record observations and issues.	
6	**Disassemble SWIFT** Including foam moulds, chassis components, conveyor, winch, electrical enclosure, and control box.	
7	**Clean components** Clean all components prior to inspection and reassembly.	

PAGE 2 OF 5

NO	ITEM	OBSERVATIONS
8	**Inspect metallic components** Inspect metallic components for corrosion, wear, or distortion.	
9	**Inspect foam moulds** Inspect foam moulds for degradation, delamination, or water ingress.	
10	**Inspect drive and tension system** Inspect drive shafts, bearings, sprockets, and tension assemblies for wear, damage, or misalignment.	
11	**Inspect chassis integrity** Inspect welds, fasteners, and chassis mounting points for damage, cracking, or deterioration.	
12	**Inspect electrical system** Inspect wiring, connectors, seals, and routing for damage, wear, or water ingress.	
13	**Replace consumables/service items** Replace consumable and service items identified as worn or corroded. Renew lubricants and grease as required.	
14	**Unspool winch rope** Fully unspool the winch rope and inspect along its full length.	
15	**Check winch rope retention** Ensure the winch rope is retained by both clamping screws.	

PAGE 3 OF 5

NO	ITEM	OBSERVATIONS
16	**Inspect winch rope** Inspect for fraying, thinning, heat damage, dirt, or oil contamination.	
17	**Inspect winch drum plating** Inspect for zinc plating loss where the rope contacts the drum side cheeks.	
18	**Grease winch** Apply lithium EP2 grease to the handle, bearings, and gears, avoiding contamination of the rope.	
19	**Check winch operation** Operate the winch and confirm smooth operation and positive holding.	
20	**Reassemble SWIFT** Reassemble the SWIFT, applying specified torque values and Loctite 243 to threaded fasteners.	
21	**Functional test** Complete a functional test and confirm baseline operation.	
22	**Inspect motor and gearbox** Inspect the motor and gearbox for lubrication leaks or signs of oil contamination.	
21	**Component inspection process** Complete the component inspection process and record findings and any actions taken.	

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NO	ITEM	OBSERVATIONS
22	**Repack and photograph for dispatch** Repackage in the shipping crate and take internal and external photos of the SWIFT and crate prior to dispatch.	

FURTHER COMMENTS

ENGINEER NAME	
ENGINEER EMAIL	

SIGNED

C UNSCHEDULED MAINTENANCE RECORD

Unscheduled maintenance may be required following a fault, damage, abnormal operation, or component failure. This record shall be completed to document the issue identified, the corrective action taken, and any parts repaired/replaced or settings changed, providing a clear maintenance trail. The SWIFT Survivor Recovery System Maintenance Manual (PN/SWI/011-00) shall be used when carrying out unscheduled maintenance.

COMPANY		LOCATION	

MODEL		SERIAL		DATE	

C.1 Unscheduled maintenance

Chapters 4 to 9 shall be used when carrying out unscheduled maintenance activities.

SUMMARY OF WORK

ENGINEER NAME	
ENGINEER EMAIL	

SIGNED

D CONTROL BOX DRAWING

The following drawing provides a visual reference of the control box, including overall dimensions and the position of mounting points and interfaces. Use this drawing to support identification, inspection, installation, and replacement activities.

TBA

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